Retention-constrained class expansion with reference-anchored candidate tables

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Abstract

A trained classifier contains discrimination that can support new-class recognition under an explicit original-class accuracy requirement. We develop reference-anchored candidate tables to make this expansion capacity measurable and usable. Fixed base references normalize learned, semantic and support-derived evidence without changing existing scores; a separate offset controls novel admission. Matched-retention frontiers quantify capacity, paired sequential comparisons measure the classifier's value, and independent protection-risk calibration selects admissible thresholds. Experiments use three datasets, ten class partitions and two frozen CLIP encoders. On ImageNet-R, retaining the classifier adds 0.71 and 0.90 percentage points over dense prototypes across five arrivals with ViT-L/14 and ViT-B/32. With up to 40 calibration images per base class, all ten partitions of each dataset and encoder certify final-session expansion and meet the 95% test-retention target. ViT-L/14 novel accuracy reaches 72.76% on CIFAR-100, 88.51% on Pets and 80.74% on ImageNet-R. A calibration-coverage condition explains when certification becomes attainable, connecting retained discrimination to

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protected admission.

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1. Introduction

An expanding recognition system already has a classifier whose decision boundaries reflect task-specific supervision. New classes arrive with names and a few examples. The central question is how much new-class recognition that existing discrimination can support while maintaining a specified accuracy on the original classes. A second question follows across repeated arrivals: how much predictive value is gained by keeping the trained classifier rather than replacing it with prototypes or recalibrated semantic evidence?

We address these questions with *reference-anchored candidate tables*. The table retains learned base columns and appends semantic anchors and support prototypes for arriving classes. Each evidence branch is normalized against fixed base references, so adding candidates leaves existing scores unchanged. An independent admission offset controls competition between base and novel classes. This separates the discrimination already learned from the decision to admit new alternatives.

The contribution is the connection between retained discrimination and class-expansion capacity: a persistent score frame makes the original classifier reusable, a common base-accuracy constraint makes its admission value comparable, and paired arrival sequences measure its continuing benefit. Independent protection-risk calibration adds a confidence-qualified admission rule and an explicit class-coverage requirement. On ImageNet-R, the gain

over dense prototypes persists across two frozen encoders. With sufficient independent calibration coverage, the tables certify final-session expansion on every evaluated partition of all three datasets and both encoders.

2. Related work

Long-tailed recognition addresses unequal class supervision through balanced fitting and losses [1, 2, 3]. CLIP supplies semantic image–text evidence [4], while prompt adaptation [5, 6], prototypes [7, 8] and semantic caches [9] adapt pre-trained representations to labeled data. Our setting combines these evidence types according to their availability at class arrival.

Class-incremental and few-shot incremental learning study retention across successive tasks [10, 11]. SimpleCIL [12] and RanPAC [13] show the value of pre-trained prototypes and analytic readouts. KBS approaches develop multi-feature similarity supplementation [14], sharpness-aware prototype calibration [15], adaptive exploration [16], representation-enhanced analytic learning [17], and multimodal prompt fusion [18]. CLIP-guided discovery [19] additionally uses unlabeled novel data, whereas our arriving classes provide names and labeled support. FeNeC [20] supplies a recent clustered-representative readout for controlled frozen-feature comparisons.

BiMC [21] and HyCal [22] calibrate multimodal or heterogeneous recognition metrics. Calibrated seen/unseen recognition [23] studies the related operating trade-off, and Learn then Test [24] supplies risk-controlled threshold selection. We focus on the retained classifier as a source of admission capacity: its contribution is measured at a common base requirement, over successive arrivals, and under a separately calibrated protection policy.

3. From retained discrimination to novel admission

3.1. Task and evidence

Base classes $\mathcal{B}$ have labeled, potentially imbalanced fitting data. Arriving classes $\mathcal{N}$ provide a name and k labeled support images per class. Prediction selects from $\mathcal{B} \cup \mathcal{N}$ without a group oracle. A frozen encoder produces a normalized feature $v = f(x)$; normalized text anchors t_c average eight fixed prompts. Learned base scores are $q_L(x, b) = v^\top w_b + b_b$, fitted by inverse-frequency-weighted ridge regression. Novel prototypes are normalized support means p_n, giving $q_P(x, n) = v^\top p_n$; semantic scores are $q_T(x, c) = v^\top t_c$.

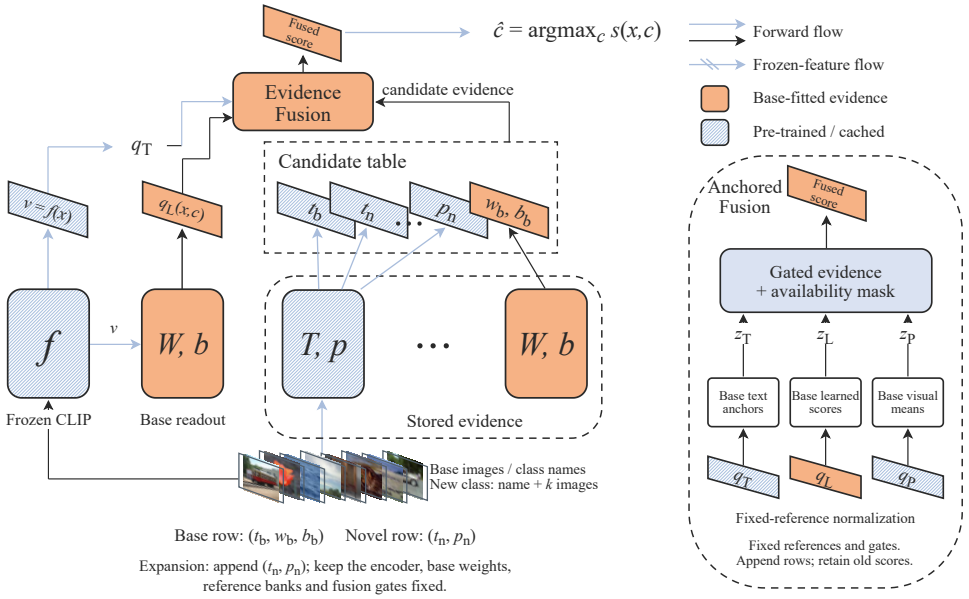


Figure 1: Reference-anchored candidate tables. Fixed base references set each branch's score frame; learned base columns remain stored as novel text anchors and support prototypes are appended. The separate novel offset controls admission. Orange marks base-fitted evidence; blue hatching marks pre-trained or cached evidence.

For each evidence source e, a fixed base reference bank $\mathcal{R}_e$ defines

$$\mu_e(x) = |\mathcal{R}_e|^{-1} \sum_{b \in \mathcal{R}_e} q_e(x, b), \tag{1}$$

$$\sigma_e^2(x) = |\mathcal{R}_e|^{-1} \sum_{b \in \mathcal{R}_e} (q_e(x, b) - \mu_e(x))^2. \tag{2}$$

We use base text anchors, learned columns and visual means as the text, learned and prototype references. Normalized evidence is

$$\widehat{q}_e(x, c) = \frac{q_e(x, c) - \mu_e(x)}{\max(\sigma_e(x), 10^{-6})}. \tag{3}$$

The statistics adapt to each query while remaining independent of the active novel vocabulary. Fusion uses the evidence available to each row:

$$s(x, b) = \widehat{q}_T(x, b) + a\widehat{q}_L(x, b), \qquad b \in \mathcal{B}, \tag{4}$$

$$s(x, n) = \widehat{q}_T(x, n) + u\widehat{q}_P(x, n) + d, \qquad n \in \mathcal{N}. \tag{5}$$

The coefficients a, u combine evidence; d controls novel admission (Fig. 1). Candidate-wise normalization instead changes the branch scales when new rows arrive, which can alter existing fused decisions. A numerical example appears in Supplementary Section K.

Proposition 1. *With fixed input, encoder, existing row evidence, reference banks, coefficients and tie-breaking, appending rows preserves every existing score and its within-vocabulary ordering.*

All terms of an existing row remain fixed. This structural property also holds for raw and fixed-temperature controls; their recognition comparisons isolate reference normalization. Appended rows can still win the expanded

prediction. Supplementary Section E gives the proof and prediction consequence. Each new row costs $O(D)$ storage and $O(kD)$ prototype construction after encoding; scoring K classes costs $O(KD)$.

3.2. Calibration and expansion capacity

A base-only pseudo-expansion episode splits base classes into pseudo-base and pseudo-novel groups. A classifier fitted on pseudo-base classes is combined with names and reserved support for pseudo-novel classes. Held-out base queries select a, u, d by harmonic macro accuracy,

$$H = 2A_B A_N/(A_B + A_N), \tag{6}$$

with zero assigned when both accuracies vanish. The coefficients then remain fixed across actual arrivals. Actual novel query labels and test labels are excluded from this selection; grids and split seeds are in Supplementary Section K.

Let $A_{B,\mathrm{ref}}^{\mathrm{closed}}$ denote the retained table's closed-base macro accuracy. At retention fraction r, every readout receives the same absolute base target. Its empirical expansion capacity is

$$N_r = \max_d A_N(d) \quad \text{subject to} \quad A_B(d) \geq r A_{B,\mathrm{ref}}^{\mathrm{closed}}. \tag{7}$$

A test-label sweep characterizes this descriptive frontier; an operational threshold uses calibration data. The exact margin envelope is derived in Supplementary Section D. Reporting these two evaluations separately distinguishes attainable capacity from selected-policy behavior.

3.3. Independent protection-risk calibration

Let R indicate correct prediction by the retained closed-base reference and C_d correct prediction by the expanded readout. Define

$$L_d = R(1 - C_d) + (1 - r)(1 - R). \tag{8}$$

Writing $D = \Pr(R = 1, C_d = 0)$ gives $\mathbb{E}L_d = D + (1 - r)(1 - A_{B,\mathrm{ref}})$. Thus $\mathbb{E}L_d \leq 1 - r$ implies $A_B(d) \geq rA_{B,\mathrm{ref}}$. This common sufficient criterion conservatively ignores replacement readouts' recovery of reference errors.

Fitting and tuning data fix the score models and ordered candidate thresholds. Independent, class-stratified calibration observations test thresholds from conservative to permissive using weighted bounded-loss p-values and fixed-sequence testing [24]. Testing stops at the first failure. Allocating error probability $\delta/5$ per arrival controls population protection across five prespecified sessions for each method, order and partition under independent class-conditional sampling. It does not guarantee arbitrary distribution shifts or every finite test outcome. If no finite threshold passes, the policy returns the reference classifier with zero novel admission. Supplementary Section J provides the bound, assumptions and full selection rule.

4. Experiments

4.1. Protocol and comparisons

We evaluate CIFAR-100 [25], Oxford-IIIT Pets [26] and ImageNet-R [27], using frozen OpenAI CLIP ViT-L/14 and ViT-B/32 features, five-shot support, ten frequency-stratified class partitions and two five-session arrival orders. Base/novel counts are 40/40, 15/15 and 100/100; each session adds eight,

three or 20 classes. Base pools follow long-tailed sampling. ImageNet-R uses task-specific splits after removing byte-identical duplicates. Novelty is relative to task fitting, with CLIP's pre-training exposure retained. Detailed pools, preprocessing and seeds are in Supplementary Sections C, I and K.

Every comparator shares its encoder, partition and support. We compare fixed-gate classifier removal, independently recalibrated removal, dense anchored text–prototype fusion, summed/class-mean Tip-Adapter caches, and raw/fixed-temperature retained fusion. The latter uses a single base-validation scale per branch. All mixtures use base-only selection; FeNeC and earlier three-partition auxiliary studies are retained in Supplementary Sections A–H.

Orders are averaged within a partition. Means and standard deviations describe ten class assignments on fixed images. Paired differences use identical cells; constrained comparisons use jointly feasible sets. Conditional paired 95% bootstrap intervals resample partition means 20,000 times. Average incremental accuracy is the mean macro accuracy over five arrivals. These intervals characterize class-assignment variation, conditional on the fixed data and encoders.

4.2. Novel recognition at a shared base requirement

At the descriptive 95% retention budget, the ImageNet-R L/14 table reaches 84.31% novel accuracy at 86.42% base accuracy. Supplementary Section I shows the complete frontiers across encoders and datasets. A more discriminative base readout can tolerate greater novel competition while meeting the same target, turning retained discrimination into admission capacity.

Reference normalization also improves the sequential comparison against

fixed-temperature fusion on ImageNet-R by 0.94 [0.74, 1.12] percentage points with L/14. Both preserve existing scores, so this comparison measures the value of query-dependent branch scaling. Full normalization controls are in Supplementary Section I.

4.3. The classifier's value across arrivals

Table 1 quantifies retained-minus-comparator average incremental accuracy with fixed base-only selected gates and offsets. On ImageNet-R, retaining the classifier adds 0.71 [0.46, 0.92] points over dense prototypes with L/14 and 0.90 [0.69, 1.11] with B/32. Corresponding gains over class-mean caches are 1.42 [1.15, 1.62] and 1.71 [1.44, 1.98] points. The result persists across both encoders and after calibration is restored to the classifier-free alternative.

CIFAR also shows gains over recalibrated removal with both encoders; its dense-prototype intervals span zero. Pets' average incremental intervals span zero for these comparators. The gain over prototype replacement is clearest on ImageNet-R.

4.4. Turning capacity into calibrated admission

An empirical calibration threshold selects the largest offset satisfying r times the reference's calibration accuracy. We separately evaluate whether test accuracy meets r times its test reference. In the original L/14 pools, the retained 95% policy is validation-feasible in all ten partitions but meets the test target in 7/10 CIFAR, 1/10 Pets and 2/10 ImageNet-R partitions. Supplementary Section I reports all methods; Section H varies threshold-label budgets.

Table 1: Paired retained-classifier gain in average incremental accuracy, percentage points [conditional paired 95% bootstrap interval]. Two orders are averaged within each of ten partitions. Full final-session differences are in Supplementary Section I.

Encoder	Dataset	Comparator	Avg. gain [95% CI]
L/14	CIFAR-100	Recalibrated removal	3.00 [2.23, 3.75]
L/14	CIFAR-100	Dense text + prototypes	0.15 [-0.46, 0.64]
L/14	CIFAR-100	Tip-Adapter mean	1.44 [0.68, 2.10]
L/14	Pets	Recalibrated removal	-0.14 [-1.53, 0.86]
L/14	Pets	Dense text + prototypes	-0.98 [-2.52, 0.22]
L/14	Pets	Tip-Adapter mean	-0.06 [-1.76, 1.41]
L/14	ImageNet-R	Recalibrated removal	2.30 [1.95, 2.59]
L/14	ImageNet-R	Dense text + prototypes	0.71 [0.46, 0.92]
L/14	ImageNet-R	Tip-Adapter mean	1.42 [1.15, 1.62]
B/32	CIFAR-100	Recalibrated removal	2.38 [1.44, 3.34]
B/32	CIFAR-100	Dense text + prototypes	0.86 [-0.15, 1.76]
B/32	CIFAR-100	Tip-Adapter mean	0.96 [-0.04, 1.84]
B/32	Pets	Recalibrated removal	-0.44 [-1.64, 0.81]
B/32	Pets	Dense text + prototypes	-0.39 [-1.85, 1.02]
B/32	Pets	Tip-Adapter mean	-0.74 [-1.62, 0.12]
B/32	ImageNet-R	Recalibrated removal	3.07 [2.82, 3.34]
B/32	ImageNet-R	Dense text + prototypes	0.90 [0.69, 1.11]
B/32	ImageNet-R	Tip-Adapter mean	1.71 [1.44, 1.98]

For independent risk calibration, the original fitting/validation pool is repartitioned into approximately 40% fitting, 20% tuning and 40% calibration observations per class. All readouts are refitted and tuned without calibration access (Supplementary Section J). We then freeze the models, support, test archives and 101 tuning-margin threshold candidates. Class-wise calibration caps of 2, 5, 10, 20 and 40 use nested subsets, supplementing the independent pool with unused training images while excluding fitting, tuning, support and test images. Every method receives identical subsets. Three subset seeds are averaged within each partition; the full cap-40 pool is counted once. We use $r = 0.95$ and $\delta = 0.05$. These benchmarks previously informed development; this is a revised benchmark evaluation.

Class coverage determines whether certification is attainable. For macro weights $w_i = 1/(Bn_c)$ and $b = \max_i w_i$, even zero observed loss requires $1/b \geq \log(\delta/5)/\log r = 89.78$ to pass the weighted bound. The original CIFAR and Pets allocations have effective counts 80 and 45–60, respectively. Increasing tail-class coverage makes useful certification possible (Fig. 2); total query counts and the derivation are in Supplementary Section L.

At cap 40, retained tables certify final-session expansion in all ten partitions of each dataset and encoder, with all test targets met (Table 2). L/14 novel accuracy is 72.76% on CIFAR, 88.51% on Pets and 80.74% on ImageNet-R. All 20 arrival paths per dataset/encoder meet every test target. Finite admission holds throughout all CIFAR and ImageNet-R paths and 17/20 L/14 and 16/20 B/32 Pets paths; the remaining Pets paths use an early-session fallback. Full comparisons distinguish policy gains, which include fallback, from accuracy differences on jointly admitted partitions. The common protection criterion

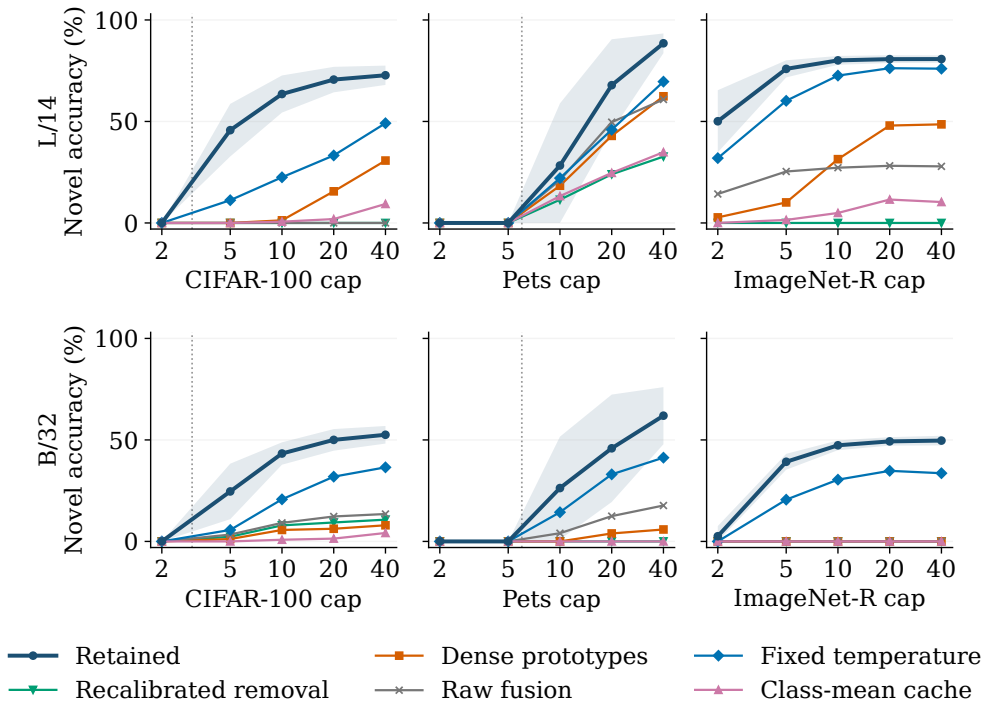


Figure 2: Novel accuracy after the fifth arrival in the first prespecified order, under independent 95% retention calibration with 95% five-arrival confidence. Means cover ten partitions, averaging three nested-subset seeds below cap 40 and the full set once at 40. Fallback contributes zero novel accuracy. Shading shows retained-table partition standard deviations, clipped to [0,100]. Grey lines mark necessary balanced class caps (CIFAR: 3; Pets: 6). ImageNet-R availability truncates the cap. All methods share the same calibration images; full counts and certification curves are in Supplementary Section L.

Table 2: Retained tables after the fifth arrival in the first prespecified order, with up to 40 independent calibration images per base class, 95% retention and 95% five-session confidence. Queries are actual base-calibration counts; Exp. counts finite certified thresholds and Test met counts attained test targets. Base/Novel are mean test percentages over ten partitions, including fallback. All six readouts are reported in Supplementary Section L.

Encoder	Dataset	Queries	Exp.	Test met	Base	Novel
L/14	CIFAR-100	1600	10/10	10/10	85.86	72.76
L/14	Pets	600	10/10	10/10	94.10	88.51
L/14	ImageNet-R	3636–3778	10/10	10/10	89.02	80.74
B/32	CIFAR-100	1600	10/10	10/10	76.74	52.54
B/32	Pets	600	10/10	10/10	88.10	61.94
B/32	ImageNet-R	3636–3778	10/10	10/10	72.98	49.67

remains conservative for replacement models that recover reference errors.

5. Conclusion

Candidate tables turn retained discrimination into reusable expansion capacity. Fixed references preserve the existing score frame, while an admission threshold exposes the trade-off between original-class protection and novel recognition. Matched-retention and paired sequential comparisons quantify this value. Independent calibration with adequate class coverage supports protected admission across three datasets and two encoders, connecting what an existing classifier has learned to how much new recognition it can support.

CRediT authorship contribution statement

Junyi Zhang: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review and editing.

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Declaration of competing interest

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The experiments use public CIFAR-100, Oxford-IIIT Pets and ImageNet-R data and the publicly released CLIP ViT-L/14 and ViT-B/32 checkpoints. Code, source-hashed experimental records, prediction archives, manuscript sources, and table/figure generation scripts are available at `https://github.com/Cheerssom/Reference-Anchored-Candidate-Tables`. Dataset images, pretrained weights, and feature caches are obtained separately using the repository instructions.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

OpenAI Codex was used to assist with literature discovery, manuscript drafting and revision, experiment and verification scripts, and LaTeX and table preparation. The author is responsible for the accuracy, interpretation and originality of the submitted work and for reviewing and approving the final manuscript and supporting materials.

References

[1] Z. Liu, Z. Miao, X. Zhan, J. Wang, B. Gong, S. X. Yu, Large-Scale Long-Tailed Recognition in an Open World, arXiv preprint arXiv:1904.05160 (2019).
URL https://arxiv.org/abs/1904.05160

[2] K. Cao, C. Wei, A. Gaidon, N. Arechiga, T. Ma, Learning Imbalanced Datasets with Label-Distribution-Aware Margin Loss, arXiv preprint arXiv:1906.07413 (2019).
URL https://arxiv.org/abs/1906.07413

[3] Y. Cui, M. Jia, T.-Y. Lin, Y. Song, S. Belongie, Class-Balanced Loss Based on Effective Number of Samples, arXiv preprint arXiv:1901.05555 (2019).
URL https://arxiv.org/abs/1901.05555

[4] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, I. Sutskever,

Learning Transferable Visual Models From Natural Language Supervision, in: International Conference on Machine Learning, 2021, pp. 8748–8763.
URL https://proceedings.mlr.press/v139/radford21a.html

[5] K. Zhou, J. Yang, C. C. Loy, Z. Liu, Learning to Prompt for Vision-Language Models, International Journal of Computer Vision 130 (9) (2022) 2337–2348. doi:10.1007/s11263-022-01653-1.
URL https://link.springer.com/article/10.1007/s11263-022-01653-1

[6] J.-X. Shi, T. Wei, Z. Zhou, J.-J. Shao, X.-Y. Han, Y.-F. Li, Long-Tail Learning with Foundation Model: Heavy Fine-Tuning Hurts, in: International Conference on Machine Learning, 2024, pp. 45014–45039.
URL https://proceedings.mlr.press/v235/shi24g.html

[7] J. Snell, K. Swersky, R. S. Zemel, Prototypical Networks for Few-shot Learning, arXiv preprint arXiv:1703.05175 (2017).
URL https://arxiv.org/abs/1703.05175

[8] Y. Wang, W.-L. Chao, K. Q. Weinberger, L. van der Maaten, SimpleShot: Revisiting Nearest-Neighbor Classification for Few-Shot Learning, arXiv preprint arXiv:1911.04623 (2019).
URL https://arxiv.org/abs/1911.04623

[9] R. Zhang, W. Zhang, R. Fang, P. Gao, K. Li, J. Dai, Y. Qiao, H. Li, Tip-Adapter: Training-free Adaption of CLIP for Few-shot Classification, arXiv preprint arXiv:2207.09519 (2022).
URL https://arxiv.org/abs/2207.09519

[10] S.-A. Rebuffi, A. Kolesnikov, G. Sperl, C. H. Lampert, iCaRL: Incremental Classifier and Representation Learning, arXiv preprint arXiv:1611.07725 (2016).
URL https://arxiv.org/abs/1611.07725

[11] X. Tao, X. Hong, X. Chang, S. Dong, X. Wei, Y. Gong, Few-Shot Class-Incremental Learning, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020, pp. 12183–12192.
URL https://openaccess.thecvf.com/content_CVPR_2020/html/Tao_Few-Shot_Class-Incremental_Learning_CVPR_2020_paper.html

[12] D.-W. Zhou, Z.-W. Cai, H.-J. Ye, D.-C. Zhan, Z. Liu, Revisiting Class-Incremental Learning with Pre-Trained Models: Generalizability and Adaptivity are All You Need, International Journal of Computer Vision (2024).
URL https://github.com/LAMDA-CL/RevisitingCIL

[13] M. D. McDonnell, D. Gong, A. Parvaneh, E. Abbasnejad, A. van den Hengel, RanPAC: Random Projections and Pre-trained Models for Continual Learning, in: Advances in Neural Information Processing Systems, Vol. 36, 2023, pp. 12022–12053. doi:10.52202/075280-0526.
URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/2793dc35e14003dd367684d93d236847-Abstract-Conference.html

[14] X. Xu, S. Niu, Z. Wang, W. Guo, L. Jing, H. Yang, Multi-feature space similarity supplement for few-shot class incremental learning, Knowledge-

Based Systems 265 (2023) 110394. doi:10.1016/j.knosys.2023.110394.

[15] R. Chen, X.-Y. Jing, F. Wu, H. Chen, Sharpness-aware gradient guidance for few-shot class-incremental learning, Knowledge-Based Systems 299 (2024) 112030. doi:10.1016/j.knosys.2024.112030.

[16] C. Han, Z. Gu, C. Xu, Z. Cui, Adaptive exploration for few-shot incremental learning, Knowledge-Based Systems 330 (2025) 114496. doi:10.1016/j.knosys.2025.114496.

[17] R. He, D. Fang, Y. Chen, K. Tong, C. Chen, Y. Wang, L.-p. Chau, H. Zhuang, REAL: Representation enhanced analytic learning for exemplar-free class-incremental learning, Knowledge-Based Systems 332 (2026) 114901. doi:10.1016/j.knosys.2025.114901.

[18] H. Fang, H. Cai, K. Mao, J. Zhao, X. Xiao, Few-shot class-incremental learning based on prompt guidance and multimodal fusion, Knowledge-Based Systems 342 (2026) 115862. doi:10.1016/j.knosys.2026.115862.

[19] Q. Yan, Y. Yang, Y. Dai, X. Zhang, K. Wiltos, M. Woźniak, W. Dong, Y. Zhang, CLIP-guided continual novel class discovery, Knowledge-Based Systems 310 (2025) 112920. doi:10.1016/j.knosys.2024.112920.

[20] K. Książek, H. Jastrzębski, K. Pniaczek, B. Trojan, M. Karp, J. Tabor, FeNeC: Enhancing continual learning via feature clustering with neighbor- or logit-based classification, Knowledge-Based Systems 338 (2026) 115479. doi:10.1016/j.knosys.2026.115479.

[21] Y. Chen, T. Ding, L. Wang, J. Huo, Y. Gao, W. Li, Enhancing few-shot class-incremental learning via training-free bi-level modality calibration, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2025, pp. 9881–9890.
URL https://openaccess.thecvf.com/content/CVPR2025/html/Chen_Enhancing_Few-Shot_Class-Incremental_Learning_via_Training-Free_Bi-Level_Modality_Calibration_CVPR_2025_paper.html

[22] E. Lee, M. Kim, J. Kwon, Y. Lee, J. Kim, S. Jang, Y. Kim, Hycal: A training-free prototype calibration method for cross-discipline few-shot class-incremental learning, arXiv:2604.15678v2 (2026).
URL https://arxiv.org/abs/2604.15678

[23] W.-L. Chao, S. Changpinyo, B. Gong, F. Sha, An Empirical Study and Analysis of Generalized Zero-Shot Learning for Object Recognition in the Wild, arXiv preprint arXiv:1605.04253 (2016).
URL https://arxiv.org/abs/1605.04253

[24] A. N. Angelopoulos, S. Bates, E. J. Candès, M. I. Jordan, L. Lei, Learn then test: Calibrating predictive algorithms to achieve risk control, arXiv preprint arXiv:2110.01052 (2021). doi:10.48550/arXiv.2110.01052.

[25] A. Krizhevsky, Learning multiple layers of features from tiny images, Tech. rep., University of Toronto (2009).
URL https://www.cs.toronto.edu/~kriz/learning-features-2009-TR.pdf

[26] O. M. Parkhi, A. Vedaldi, A. Zisserman, C. V. Jawahar, Cats and dogs,

in: IEEE Conference on Computer Vision and Pattern Recognition, 2012, pp. 3498–3505. doi:10.1109/CVPR.2012.6248092.
URL https://www.robots.ox.ac.uk/~vgg/publications/2012/parkhi12a/

[27] D. Hendrycks, S. Basart, N. Mu, S. Kadavath, F. Wang, E. Dorundo, R. Desai, T. Zhu, S. Parajuli, M. Guo, D. Song, J. Steinhardt, J. Gilmer, The Many Faces of Robustness: A Critical Analysis of Out-of-Distribution Generalization, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 8340–8349.
URL https://openaccess.thecvf.com/content/ICCV2021/html/Hendrycks_The_Many_Faces_of_Robustness_A_Critical_Analysis_of_Out-of-Distribution_ICCV_2021_paper.html

Supplementary material: Retention-constrained class expansion with reference-anchored candidate tables

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Sections A–H retain the original three-partition ViT-L/14 auxiliary and validation studies. Section I reports the ten-partition, two-encoder comparison; Section J gives the independent protection bound and original calibration allocation; Section K collects implementation details; Section L evaluates independent calibration coverage with frozen score models. Their statistical units are stated separately.

Appendix A. Auxiliary readout and mechanism analyses

For the auxiliary readout and support-budget comparisons, the gate grid is $a, u \in \{0, 0.25, 0.5, 1, 2\}$ and $d \in \{-2, -1.5, \ldots, 2\}$. Offset-calibrated controls use their separately stated grids. The main retention/value evaluation uses the common expanded grid in Section 3.2. All experiments share the same frozen representations, true class partitions and disjoint support pools.

Appendix A.1. Readouts and evaluation measures

All compared readouts use the same CLIP image embeddings, text anchors, class partitions, and novel support images. The text baseline uses cosine similarity to class names. The visual prototype baseline uses means of base fitting examples and means of novel support examples. Text–prototype fusion adds the two cosine branches with a mixing coefficient selected on the same

pseudo-novel calibration episode; its grid is $\{0, 0.25, 0.5, 1, 2, 4\}$, including pure text.

Tip-Adapter uses a balanced k-image cache per candidate and scores $100q_T + \alpha q_{\text{cache}}$, with affinity $\exp[\beta(v^\top f(x_i) - 1)]$. The same base-only episode selects $\alpha \in \{0, 0.5, 1, 2, 4\}$ and $\beta \in \{1, 2, 5, 10\}$. Cache images come from the reserved pool for both groups; base-trained readouts use the full base fitting set.

We also compare raw evidence fusion and candidate-normalized fusion. Raw fusion uses unstandardized text, learned, and prototype scores with the same coefficient grid as the proposed table. Candidate-normalized fusion standardizes text over all active classes, learned scores over the base classes, and prototypes over active novel classes. It shares the base classifier and calibration protocol with reference-anchored fusion. This comparison changes the normalization population while holding the principal evidence sources fixed.

We report macro base accuracy A_B, macro novel accuracy A_N, all-class macro accuracy, and the harmonic mean in Eq. (6). Both datasets use equal numbers of base and novel classes, so all-class macro accuracy is $(A_B + A_N)/2$. On the balanced CIFAR test set this also equals sample accuracy; on Pets the macro measure gives each breed equal weight. Stability measures include maximum absolute change in an existing score, change of the maximizing class restricted to base classes, and prediction changes over the expanded vocabulary. Error flips are reported as fractions of all base test images, not conditional fractions of previously correct images.

Table S1: Five-shot expansion from 40 base classes to 80 classes. All values are accuracy percentages, reported as mean ± sample standard deviation over three distinct class partitions. All readouts share frozen CLIP features; coefficients are selected using base-only calibration. Base and novel performance jointly specify each operating point.

Readout	Base	Novel	All	Harmonic
CLIP text	78.32 ± 0.25	76.86 ± 1.97	77.59 ± 0.86	77.57 ± 0.88
Visual prototypes	77.53 ± 2.03	60.92 ± 3.94	69.23 ± 2.09	68.17 ± 2.49
Text + prototypes	82.47 ± 0.87	77.53 ± 0.47	80.00 ± 0.56	79.92 ± 0.55
Tip-Adapter (tuned)	78.38 ± 0.24	77.42 ± 1.72	77.90 ± 0.83	77.89 ± 0.83
Raw evidence fusion	78.32 ± 0.25	76.86 ± 1.97	77.59 ± 0.86	77.57 ± 0.88
Candidate-normalized table	85.36 ± 0.66	70.83 ± 4.33	78.09 ± 2.02	77.36 ± 2.48
Reference-anchored table	86.03 ± 1.40	66.90 ± 4.23	76.46 ± 1.43	75.19 ± 2.14

Appendix A.2. Recognition after five-shot class expansion

Reference-anchored fusion reaches 86.03 ± 1.40 base accuracy and 66.90 ± 4.23 novel accuracy (Table S1). Its base accuracy is 7.71 percentage points above text-only classification and 3.56 points above text–prototype fusion, providing a retention-oriented operating point for class expansion.

Text–prototype fusion uses visual evidence for every class and reaches 80.00 ± 0.56 all-class accuracy and 77.53 ± 0.47 novel accuracy. Candidate-normalized fusion reaches 85.36 ± 0.66 base and 70.83 ± 4.33 novel accuracy. These points characterize the recognition trade-off accompanying the reference-anchored score system.

Raw fusion selects the text-only point in all five-shot CIFAR partitions. Both normalization rules activate learned evidence in their selected solutions, demonstrating the role of scale alignment when combining classifier logits with cosine similarities.

Appendix A.3. Fine-grained expansion on Oxford-IIIT Pets

On fine-grained Pets expansion, reference anchoring reaches 97.18 ± 2.25 base and 87.79 ± 6.18 novel accuracy (Table S2). Base accuracy is 2.41

Table S2: Five-shot Oxford-IIIT Pets expansion from 15 to 30 breeds under long-tailed base supervision. Values are macro accuracy percentages, mean ± sample standard deviation across three distinct breed partitions. Grids and the base-only calibration rule are transferred from the CIFAR experiment without dataset-specific search expansion.

Readout	Base	Novel	All	Harmonic
CLIP text	92.87 ± 3.37	94.05 ± 1.35	93.46 ± 1.59	93.43 ± 1.63
Visual prototypes	92.08 ± 2.61	68.30 ± 4.62	80.19 ± 2.33	78.35 ± 2.97
Text + prototypes	94.77 ± 2.91	93.91 ± 1.55	94.34 ± 1.43	94.31 ± 1.44
Tip-Adapter (tuned)	93.56 ± 2.45	94.07 ± 1.18	93.81 ± 1.27	93.80 ± 1.28
Raw evidence fusion	90.13 ± 8.01	91.18 ± 4.27	90.66 ± 6.13	90.63 ± 6.18
Candidate-normalized table	96.85 ± 1.50	86.49 ± 7.64	91.67 ± 4.56	91.29 ± 4.99
Reference-anchored table	97.18 ± 2.25	87.79 ± 6.18	92.49 ± 2.83	92.14 ± 3.23

Table S3: Additional five-shot controls with the same frozen encoder, support images, and base-only calibration. Values are mean ± sample standard deviation across three partitions. The random-projection readout is a controlled frozen RanPAC-style adaptation described in the text.

CIFAR-100

Readout	Base	Novel	All
Text + group offset	78.93 ± 2.75	75.56 ± 4.40	77.24 ± 0.89
Anchored text + prototypes	82.40 ± 2.02	76.19 ± 3.18	79.30 ± 0.59
Frozen RP-ridge (2000)	70.18 ± 0.69	52.52 ± 4.27	61.35 ± 1.83

Oxford-IIIT Pets

Readout	Base	Novel	All
Text + group offset	95.29 ± 3.80	89.85 ± 4.86	92.57 ± 2.31
Anchored text + prototypes	94.58 ± 3.22	90.38 ± 5.41	92.48 ± 1.10
Frozen RP-ridge (2000)	93.49 ± 3.32	73.70 ± 5.13	83.60 ± 1.13

points above text–prototype fusion. Mean base/novel accuracy also increases from 96.85%/86.49% with candidate normalization to 97.18%/87.79% with reference anchoring.

Text–prototype fusion reaches 94.34 ± 1.43 all-class macro accuracy; the reference-anchored table reaches 92.49 ± 2.83 with greater retention of base-breed discrimination. Both datasets use the same search grids and base-only calibration rule.

Appendix A.4. Calibration controls and analytic incremental readouts

Offset-calibrated controls share the same pseudo-novel episode (Table S3). Text + group offset uses base-referenced text scores and a novel offset. Anchored text + prototypes uses fixed base references for both branches and

visual evidence for every class. Harmonic selection searches offsets from -4 to 4 in steps of 0.25 and prototype weights $\{0, 0.25, 0.5, 1, 2, 4\}$. Fixed-reference text–prototype fusion also satisfies Proposition 1; the score invariant applies across availability masks.

Anchored text–prototype fusion attains 82.40% base and 76.19% novel accuracy on CIFAR, and 94.58%/90.38% on Pets. Offset-calibrated text attains 78.92%/75.56% and 95.29%/89.85%, respectively. Together with the heterogeneous table, these controls separate group calibration, evidence configuration, and the shared fixed-reference property.

The analytic baseline follows the random-projection phase of RanPAC [1]. A fixed Gaussian matrix maps CLIP features to 2000 ReLU coordinates, and $(G + \lambda I)W = Q$ accumulates Gram matrices and class cross-products from base fitting data and novel support. Base validation macro accuracy selects $\lambda \in \{10^{-8}, \ldots, 10^{8}\}$; the pseudo episode selects a novel offset. This frozen adaptation uses base validation in place of the original within-task MSE selection. It attains 70.18%/52.53% base/novel accuracy on CIFAR and 93.49%/73.70% on Pets. Novel statistics refit the analytic readout; candidate-table expansion appends rows to its fixed readout.

Under the common 95% validation retention target, reference anchoring reaches 84.56% base and 73.87% novel test accuracy on CIFAR, compared with 82.84%/77.11% for anchored text–prototype fusion (Table S4). Both are feasible on all three validation partitions. On Pets, the corresponding test points are 97.18%/87.79% and 94.03%/90.96%. Text alone reaches the CIFAR validation target in one partition. Targets constrain validation selection; the table reports the resulting test operating points. Complete 95% and 99%

Table S4: Validation retention-targeted calibration at five shots. The pseudo-base target is 95% of the closed reference-table pseudo-base accuracy. Among feasible grid points, selection maximizes pseudo-novel accuracy; infeasible partitions use the point with highest pseudo-base accuracy, then pseudo-novel accuracy. Feasible counts refer to validation partitions. Test values are percentages, mean ± sample standard deviation; test retention is not constrained.

Data	Readout	Base	Novel	All	Feasible
CIFAR	Text + group offset	81.47 ± 1.13	71.49 ± 5.03	76.48 ± 1.99	1/3
CIFAR	Anchored text + prototypes	82.84 ± 1.50	77.11 ± 4.46	79.98 ± 1.61	3/3
CIFAR	Candidate-normalized table	84.69 ± 1.92	72.45 ± 4.14	78.57 ± 1.54	3/3
CIFAR	Reference-anchored table	84.56 ± 1.31	73.88 ± 5.01	79.22 ± 2.21	3/3
Pets	Text + group offset	95.29 ± 3.80	89.85 ± 4.86	92.57 ± 2.31	3/3
Pets	Anchored text + prototypes	94.03 ± 3.37	90.96 ± 5.91	92.50 ± 1.27	3/3
Pets	Candidate-normalized table	97.25 ± 1.78	85.26 ± 7.25	91.26 ± 4.40	3/3
Pets	Reference-anchored table	97.18 ± 2.25	87.79 ± 6.18	92.49 ± 2.83	3/3

results accompany the experimental records.

Appendix A.5. Vocabulary growth and score preservation

The vocabulary test appends nested sets of 10, 20, and 40 novel classes to the 40 CIFAR base classes, using sorted novel IDs and fixed five-shot gates. At each stage, we compare existing scores with the original base decision.

Reference anchoring gives zero measured base-score drift and zero base-only ranking churn across all nine CIFAR partition and vocabulary settings (Fig. S1(a,b)). Candidate normalization changes existing scores by up to approximately 1.29–1.74 normalized units and changes a fraction of base-only predictions. The Pets test appends 3, 7, and 15 breeds and likewise records zero anchored score drift and ranking churn in all nine conditions.

At 40 appended classes, expanded predictions change on $3.63 \pm 2.40\%$ of base images, including $1.84 \pm 1.54\%$ correct-to-wrong changes (Fig. S1(c)). Wrong-to-correct changes are zero, as specified by Corollary 1. The preserved base scores isolate these changes as competition from appended rows.

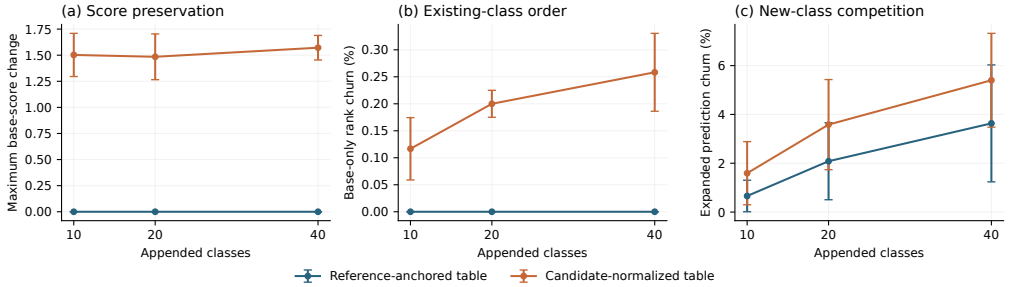


Figure S1: Vocabulary growth with five-shot novel prototypes. Each curve averages three true class partitions; error bars show sample standard deviation. Reference-anchored fusion preserves existing scores and base-only rankings. Expanded predictions can still change when a new candidate wins. Maximum score drift is measured across all test queries and existing base classes.

Table S5: Reference-anchored expansion across novel support budgets. Accuracy percentages are aggregated across partitions after averaging support subsets within each partition.

Shots	Base	Novel	All	Harmonic
1	84.57 ± 0.88	70.86 ± 5.41	77.72 ± 2.27	77.00 ± 2.85
2	85.65 ± 1.45	70.05 ± 1.13	77.85 ± 0.67	77.04 ± 0.67
5	86.03 ± 1.40	66.90 ± 4.23	76.46 ± 1.43	75.19 ± 2.14

Five-session expansion adds eight CIFAR classes or three Pets breeds per session, with two random arrival orders per partition. Across 30 session records per dataset, comparisons of all previously admitted rows, including novel rows, give maximum floating-point score differences of 2.85×10^{-5} on CIFAR and 3.58×10^{-6} on Pets. Both arrival orders produce identical final predictions in every partition under class-ID tie breaking.

Appendix A.6. Support budget and evidence availability

At one, two, and five shots, base accuracy remains between 84.57% and 86.03%, while novel accuracy ranges from 66.90% to 70.86% (Table S5; Fig. S2). Calibration includes zero branch coefficients, allowing the selected

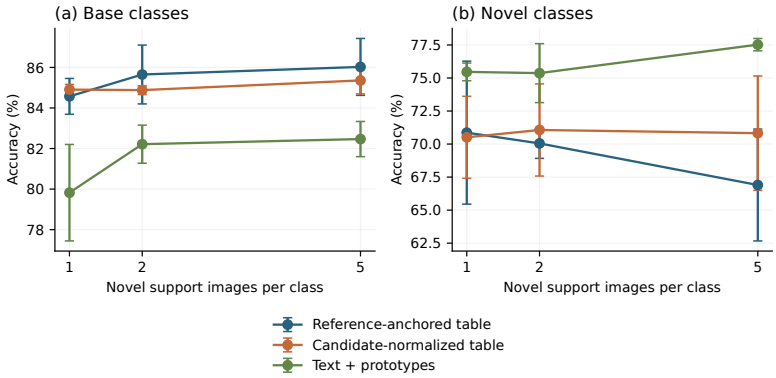


Figure S2: Base and novel accuracy at one, two, and five support images per novel class. The same fixed-reference scoring rule is used at every budget, with coefficients selected on the corresponding base-only pseudo episode. Error bars represent variation between class partitions.

Table S6: Evidence removal at the selected five-shot reference-anchored operating point, with all remaining coefficients fixed. Accuracy percentages are mean ± sample standard deviation over three class partitions.

Readout	Base	Novel	All	Harmonic
Reference-anchored table	86.03 ± 1.40	66.90 ± 4.23	76.46 ± 1.43	75.19 ± 2.14
Without learned evidence	58.10 ± 32.21	80.72 ± 4.47	69.41 ± 13.87	62.85 ± 25.08
Without novel prototypes	86.42 ± 1.71	47.17 ± 30.85	66.79 ± 14.62	56.15 ± 30.05
Without offset	86.14 ± 1.69	65.58 ± 7.35	75.86 ± 2.83	74.25 ± 4.08

mixture to adapt to the support budget. Each budget defines a separately calibrated operating point: support determines the prototype evidence, and fusion determines its competition with learned base discrimination.

Appendix A.7. Mechanism ablation and operating trade-off

Removing learned evidence, novel prototypes, or the offset changes a different part of the selected decision rule (Table S6). With the remaining coefficients fixed, these interventions identify how each source contributes to the five-shot operating point. The independently calibrated text and text–prototype readouts provide the corresponding reduced-model comparisons.

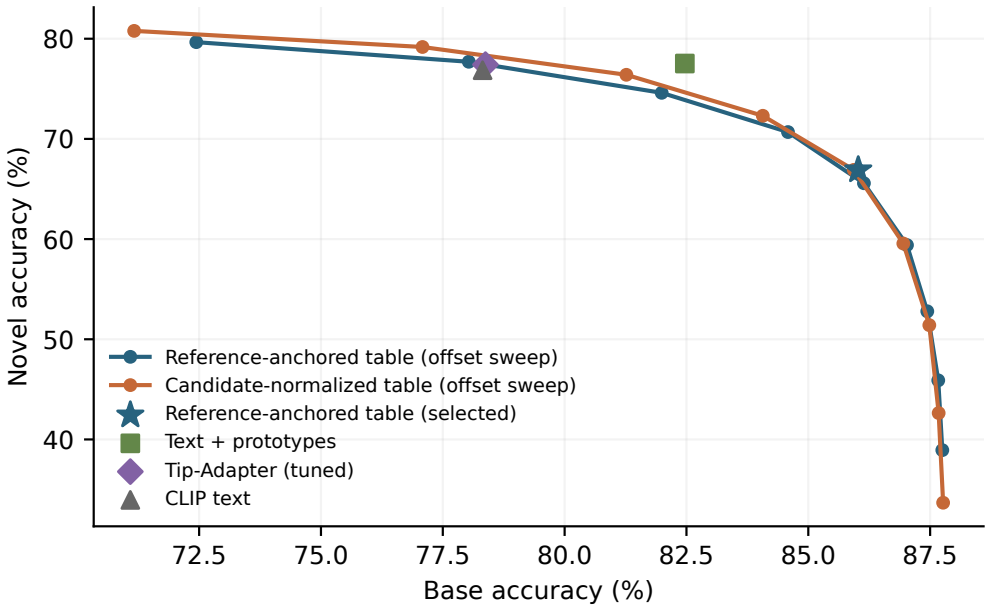


Figure S3: Base–novel operating curves for five-shot expansion. The offset is swept with the other selected coefficients fixed; these test curves are descriptive analyses and are not used to select the reported operating points. Symbols identify base-only calibrated points and matched baselines.

Figure S3 sweeps d from -2 to 2 in steps of 0.5 with a and u fixed. Increasing d shifts the margin between the best base and novel candidates while preserving within-group rankings (Eq. (5)). The resulting curve identifies how the offset controls admission of new categories; a validation retention target selects the operating point.

Appendix A.8. Readout efficiency

Candidate scoring from cached 768-dimensional features takes approximately 1.76, 0.75, and 0.78 ms per batch for 50, 60, and 80 candidates. Measurements use batches of 256 on an NVIDIA RTX 4070 Laptop GPU, five warm-up iterations, and 50 timed iterations with GPU synchronization.

They include reference statistics and fusion and exclude image/text encoding. At these vocabulary sizes, launch and reduction overheads dominate the measured latency.

Appendix B. Validation-selected retention policy

Table S7 applies the common validation target after five arrivals. Raw records include all five target fractions and every arrival session. Infeasible targets are reported without substituting a zero novel accuracy. Comparisons with different feasible counts refer to different partition sets.

Table S7: Final-session test accuracies after the admission threshold is selected using base validation only. Budgets use the same closed-base validation reference for all methods within a partition. Values are percentages, mean ± sample standard deviation over feasible partitions; counts identify those partitions. Test accuracies are the outcomes of this operational policy and are not matched by test labels.

Data	Budget	Readout	Base	Novel	Feasible
CIFAR	90%	Retained classifier	78.33 ± 3.89	80.13 ± 2.25	3/3
CIFAR	90%	Classifier removed (fixed gates)	79.07 ± 3.94	70.79 ± 13.81	3/3
CIFAR	90%	No classifier (recalibrated)	78.73 ± 3.21	75.45 ± 5.43	3/3
CIFAR	90%	Anchored text + prototypes	80.18 ± 0.60	79.45 ± 0.76	3/3
CIFAR	95%	Retained classifier	82.99 ± 2.29	75.71 ± 3.14	3/3
CIFAR	95%	Classifier removed (fixed gates)	81.95 ± 0.04	71.54 ± 3.87	2/3
CIFAR	95%	No classifier (recalibrated)	81.85 ± 0.18	71.70 ± 4.10	2/3
CIFAR	95%	Anchored text + prototypes	81.95 ± 0.64	77.48 ± 1.52	2/3
CIFAR	99%	Retained classifier	86.48 ± 1.65	62.83 ± 10.11	3/3
CIFAR	99%	Classifier removed (fixed gates)	–	–	0/3
CIFAR	99%	No classifier (recalibrated)	–	–	0/3
CIFAR	99%	Anchored text + prototypes	86.85	53.85	1/3
Pets	90%	Retained classifier	84.87 ± 4.59	96.53 ± 0.78	3/3
Pets	90%	Classifier removed (fixed gates)	84.11 ± 2.76	96.44 ± 0.64	3/3
Pets	90%	No classifier (recalibrated)	82.00 ± 0.20	96.38 ± 0.52	3/3
Pets	90%	Anchored text + prototypes	85.69 ± 4.62	95.87 ± 1.57	3/3
Pets	95%	Retained classifier	90.29 ± 2.90	95.68 ± 0.83	3/3
Pets	95%	Classifier removed (fixed gates)	90.11 ± 1.17	95.71 ± 0.11	2/3
Pets	95%	No classifier (recalibrated)	89.88 ± 0.85	95.71 ± 0.11	2/3
Pets	95%	Anchored text + prototypes	89.37 ± 4.15	95.63 ± 1.55	3/3
Pets	99%	Retained classifier	96.43 ± 2.15	88.67 ± 7.89	3/3
Pets	99%	Classifier removed (fixed gates)	–	–	0/3
Pets	99%	No classifier (recalibrated)	–	–	0/3
Pets	99%	Anchored text + prototypes	97.70 ± 1.11	84.19 ± 12.46	2/3

Appendix C. Reproduction details

The expansion experiment is implemented in `src/experiments/fcs_expansion.py`. It can be run from any working directory. The manifest records the code hash, frozen-cache hash, image-pool hashes, class assignments, pseudo-class assignments, support indices, regularization choices, and selection grids. The frozen support features are computed with the same deterministic interpolation used for the frozen training cache. The cached image embeddings

are checked numerically before scoring.

Every novel support image comes from the five-image-per-class reserve created before long-tail sampling. This supplies a consistent support budget even for tail classes whose fitting pool contains fewer than five images. At five shots the three subset seeds select the same images in different orders, so the reported standard deviation is calculated across the three distinct class partitions. One- and two-shot results average the three subsets within each partition before computing the partition-level variation.

The main result JSON contains all compared readouts, including evidence-removal ablations. The vocabulary records contain the maximum score drift and both restricted and expanded prediction churn. The operating-curve records contain every swept offset, with no selection based on test outcomes. The efficiency records specify batch size and measured vocabulary size. The manuscript tables are generated directly from these files, and the summary CSV exposes both means and sample standard deviations.

The semantic prompts are "a photo of a {}.", "a bad photo of a {}.", "a cropped photo of a {}.", "a close-up photo of a {}.", "a bright photo of a {}.", "a good photo of a {}.", "a photo of the {}.", and a second occurrence of "a photo of a {}.". The repeated prompt follows the existing cached anchor construction and is retained consistently for all methods.

Frozen CLIP features are the sole image representation in the expansion experiments. Six fitting images are independently re-embedded and checked for cosine agreement above 0.999 with the cached features. Image hashes verify that fitting, validation, and reserved-support pools are disjoint. The fixed base text block is computed separately from novel text columns; sequential

prototype products can exhibit the reported floating-point residuals when matrix widths change.

Appendix D. Interpreting group competition

Let $M_B(x) = \max_{b \in \mathcal{B}}[\widehat{q}_T(x, b) + a\widehat{q}_L(x, b)]$ and $M_N(x) = \max_{n \in \mathcal{N}}[\widehat{q}_T(x, n) + u\widehat{q}_P(x, n)]$. The classifier selects a novel class precisely when $M_N(x) + d > M_B(x)$, subject to the fixed tie-breaking rule. Hence, the offset is a threshold on the margin $M_B - M_N$. Increasing d cannot change the winning base class or the winning novel class when all other quantities are fixed. It changes which group supplies the final winner. This explains the direction of the operating curves without assuming a calibrated probability interpretation for the fused score.

For nested candidate sets and fixed references, adding novel rows can increase M_N but cannot change M_B. An existing-class example may therefore cross the admission boundary as the vocabulary grows, while its base scores remain identical. This distinguishes the deployment behavior measured by expanded prediction churn from the normalization behavior measured by base-only ranking churn.

Appendix D.1. Exact empirical retention envelope

For a fixed readout, let b_i and n_i be the canonical winning base and novel class for query i, and let $\Delta_i = M_B(x_i) - M_N(x_i)$. Define $g_i(d) = \mathbf{1}[d > \Delta_i \text{ or } (d = \Delta_i \text{ and } n_i < b_i)]$. With the class-macro weights w_i, the two group

accuracies are

$$A_B(d) = \sum_{i:y_i \in \mathcal{B}} w_i \mathbf{1}[b_i = y_i](1 - g_i(d)), \tag{D.1}$$

$$A_N(d) = \sum_{i:y_i \in \mathcal{N}} w_i \mathbf{1}[n_i = y_i] g_i(d). \tag{D.2}$$

Thus A_B is nonincreasing and A_N is nondecreasing in d. For the stored finite-precision scores, an exact search need only consider each observed margin and its immediately preceding representable offset, together with the all-base and all-novel endpoints. Between adjacent margins every group decision is fixed. If τ exceeds $\sum_{i:y_i \in \mathcal{B}} w_i \mathbf{1}[b_i = y_i]$, the constraint is infeasible. Otherwise, sorting the margins of correctly classified base queries in descending order and accumulating their macro weights locates the boundary at which retained base mass first reaches τ. The tie rule selects that boundary or its predecessor. This is the largest feasible offset in the implementation's offset ordering. Monotonicity of A_N proves that it maximizes novel accuracy under the empirical constraint. The argument applies identically to base-validation selection; that policy's test outcome is evaluated on separate queries.

The envelope separates the within-base feasible ceiling, the admitted group threshold and the distribution of correctly classified novel-query margins. It therefore measures the joint operating value of a learned readout and its competition with arriving evidence, beyond the statement that stored scores are preserved.

Appendix E. Additional background and score-preservation proof

Appendix E.1. Long-tailed recognition and calibration

Long-tailed recognition addresses the bias induced by unequal class frequencies. Margin adjustment allocates different classification margins to

frequent and rare classes [2]; class-balanced weighting uses effective sample counts [3]; and prior-aware objectives modify the relationship between logits and class probabilities [4, 5]. Representation–classifier decoupling provides a complementary route: a representation is learned first and a balanced classifier is then fitted to that representation [6]. These developments motivate our use of a balanced base classifier as one source of evidence.

Foundation models change the representation-learning budget and the role of task-specific adaptation. LIFT studies lightweight adaptation of pre-trained models for long-tail learning [7]. Low-rank updates provide an efficient mechanism for adapting large parameter matrices [8], including transformer backbones [9]. Our candidate interface can accept weights obtained through these routes. The expansion experiments here fix the encoder throughout and isolate the effect of the readout.

Long-tailed calibration studies confidence under class imbalance [10], while robust long-tailed learning addresses its interaction with label noise [11]. We study the reference population used to standardize evidence as candidate classes change. Class-balanced validation and separate base/novel metrics expose how the resulting score system allocates discrimination.

Appendix E.2. Semantic and visual evidence for few-shot recognition

CLIP constructs classifiers from natural-language supervision and supports classification by changing the text candidates [12]. CoOp learns prompt context for target tasks, while CoCoOp introduces conditional prompting to improve transfer across categories [13, 14]. These methods concern the construction or adaptation of semantic evidence. Our table keeps semantic anchors available to every class and combines them with evidence that may

exist for only a subset of the vocabulary.

Prototype methods summarize class examples in an embedding space [15]. SimpleShot studies feature normalization for nearest-neighbor classification [16], and Tip-Adapter augments CLIP with a support-feature cache [17]. Few-shot transfer under distribution shift further motivates separating shared and task-specific knowledge [18]. Our comparisons use a shared encoder and novel support budget to examine evidence availability and its normalization reference.

Appendix E.3. Proof of the main score-preservation proposition

Proof. For an existing candidate c, each raw score $q_e(x, c)$ and mask m_{ec} is fixed. The means and variances in Eq. (2) are functions of x and $\mathcal{R}_e$, and neither depends on the appended rows. Therefore each normalized score in Eq. (3) is unchanged. The coefficients and offset of c are fixed, so their sum is unchanged. Subtracting the unchanged scores of any two existing rows gives an unchanged difference. A fixed tie-breaking rule preserves the maximizing existing row. □

The proposition covers both base rows and previously appended novel rows with fixed prototypes. The implementation computes the fixed base text block separately from novel columns to keep its numerical kernel independent of the appended width.

Corollary 1. *If every existing row has a preserved score, a changed final prediction after expansion must select an appended row. An example whose previous prediction was incorrect cannot become correct after adding classes when its true class belongs to the existing vocabulary.*

Proof. The best existing row is unchanged by Proposition 1. The expanded maximizer is either this row or a newly appended row. For an example labeled by an existing class, selecting a new row cannot correct a previous error, and keeping the existing maximizer leaves the previous error unchanged. □

Appendix F. Recent readout comparison and ImageNet-R extension

The extension protocol was recorded before ImageNet-R evaluation. Class partitions use seeds 0, 1 and 2, with exact frequency-stratified allocation of 100 base and 100 novel classes. Within each base group, seed $900 + p$ assigns 50 pseudo-base and 50 pseudo-novel classes. Base-only validation selects ridge regularization, branch mixtures and readout hyperparameters. No ImageNet-R test result selects these quantities. The retained table, removal controls and dense prototype replacement use the same grids as the CIFAR/Pets experiment.

FeNeC [19] is independently implemented from the paper's equations and checked against author-code commit `1575ea019fd583149c47c068f7f5ccc102ea4580`. Both variants use identity Tukey transformation for signed CLIP features and Euclidean k-means with a maximum of 100 iterations. The representative grid is $K \in \{1, 5, 25\}$ and the neighbor grid is $\{1, 3, 5\}$. FeNeC-M estimates one covariance per class, applies the author's ViT single-shrinkage and correlation normalization with $(\gamma_1, \gamma_2) \in \{(1, 1), (6.12, 8.10), (9.98, 0)\}$, then uses inverse squared-Mahalanobis-distance votes. FeNeC-E uses inverse unsquared-Euclidean-distance votes. Their native predictions are preserved when votes are converted to log votes, with a floor of 10^{-30}. The calibrated forms select a novel offset from $[-4, 4]$ in steps of 0.25 on the pseudo episode. Harmonic macro accuracy chooses the configuration.

The five-shot protocol caps a class's representative count at its sample count. A class with one fitting sample uses identity covariance, avoiding an undefined sample covariance. These are explicit sparse-data adaptations. A

per-class independent summary can be cached before its arrival; the voting neighborhood always contains only currently admitted classes. Native and calibrated forms share each current-vocabulary score matrix, so their complete matched-retention frontiers coincide, while fixed-offset trajectories can differ. The final grids are documented with the code. FeNeC frontiers use its sparse global-neighbor votes, with unvoted classes at the common floor; feasibility refers to this output interface.

All ImageNet-R images are content-hashed before pool assignment. Fifty-nine byte-identical duplicates are removed globally. No retained content hash appears in two fitting, validation, support or test pools. The source archive has SHA-256 `18c6bf493b39a0d975d48e587437f562caab9c52ae6327dcfa9dd8eb54aa1b52`. The class-name mapping, image paths, achieved fitting counts, source hashes, calibrated settings and prediction margins accompany the code and records. These are task-specific splits of ImageNet-R, whose source release is a rendition benchmark [20].

Table S8: ImageNet-R matched-retention recognition at three budgets after five arrivals. Values are percentages, mean ± partition-level sample standard deviation over feasible partitions. Counts specify available partition sets.

Budget	Readout	Base	Novel	Feasible
90%	Retained classifier	81.83 ± 0.29	86.80 ± 0.27	3/3
90%	Classifier removed (fixed gates)	81.81 ± 0.31	85.53 ± 0.49	3/3
90%	No classifier (recalibrated)	81.81 ± 0.31	85.34 ± 0.46	3/3
90%	Anchored text + prototypes	81.83 ± 0.32	86.59 ± 0.58	3/3
90%	Tip-Adapter sum + offset	81.82 ± 0.30	83.74 ± 0.25	3/3
90%	Tip-Adapter mean + offset	81.83 ± 0.30	85.30 ± 0.23	3/3
95%	Retained classifier	86.38 ± 0.32	84.48 ± 0.72	3/3
95%	Classifier removed (fixed gates)	86.37 ± 0.31	72.62 ± 12.51	3/3
95%	No classifier (recalibrated)	86.37 ± 0.31	72.69 ± 12.57	3/3
95%	Anchored text + prototypes	86.37 ± 0.32	83.32 ± 1.40	3/3
95%	Tip-Adapter sum + offset	86.41 ± 0.30	70.93 ± 10.32	3/3
95%	Tip-Adapter mean + offset	86.38 ± 0.33	81.58 ± 1.72	3/3
99%	Retained classifier	90.02 ± 0.34	73.69 ± 1.50	3/3
99%	Classifier removed (fixed gates)	–	–	0/3
99%	No classifier (recalibrated)	–	–	0/3
99%	Anchored text + prototypes	89.98 ± 0.47	47.62 ± 13.36	2/3
99%	Tip-Adapter sum + offset	–	–	0/3
99%	Tip-Adapter mean + offset	–	–	0/3

Table S9: ImageNet-R test outcomes with the 95% base-validation-only threshold policy. Validation feasibility is reported separately from achieved test accuracy. Tip-Adapter uses its base-only selected calibrated cache settings, followed by validation-only threshold selection.

Readout	Base test	Novel test	Validation feasible
Retained classifier	86.01 ± 0.59	84.73 ± 0.71	3/3
Classifier removed (fixed gates)	86.38 ± 0.18	57.35 ± 38.51	3/3
No classifier (recalibrated)	86.48 ± 0.20	57.10 ± 38.30	3/3
Anchored text + prototypes	86.36 ± 0.47	83.26 ± 1.49	3/3
Tip-Adapter sum + offset	86.10 ± 0.46	74.08 ± 5.48	3/3
Tip-Adapter mean + offset	86.68 ± 0.23	80.75 ± 2.09	3/3

Table S10: FeNeC matched-test-retention operating points at strict budgets. Values are percentages, mean ± sample standard deviation over feasible partitions; missing values have zero feasible partitions. Native and offset-calibrated forms share a score matrix and hence the same complete offset-sweep frontier.

Data	Budget	Readout	Base	Novel	Feasible
CIFAR-100	95%	FeNeC-M + offset	–	–	0/3
CIFAR-100	95%	FeNeC-E + offset	–	–	0/3
CIFAR-100	99%	FeNeC-M + offset	–	–	0/3
CIFAR-100	99%	FeNeC-E + offset	–	–	0/3
Pets	95%	FeNeC-M + offset	–	–	0/3
Pets	95%	FeNeC-E + offset	94.23	69.46	1/3
Pets	99%	FeNeC-M + offset	–	–	0/3
Pets	99%	FeNeC-E + offset	–	–	0/3
ImageNet-R	95%	FeNeC-M + offset	–	–	0/3
ImageNet-R	95%	FeNeC-E + offset	–	–	0/3
ImageNet-R	99%	FeNeC-M + offset	–	–	0/3
ImageNet-R	99%	FeNeC-E + offset	–	–	0/3

Appendix G. Dense caches, repeated calibration and measured readout cost

Appendix G.1. Dense semantic-cache protocol

The training-free Tip-Adapter score follows the ECCV paper and the author implementation [17]: cosine similarity to labeled cache keys is transformed by $\exp[-\beta(1 - \text{similarity})]$, summed into classes and added to 100 times text similarity with weight α. The author code is pinned at commit `d0e2d6f8c5feb8b6ce937b757810761f7155d4d5`. All permitted base fitting images and five reserved novel support images supply the keys. The sum variant retains the published aggregation. A separate class-mean variant divides each class's retrieval score by its key count, isolating the effect of unequal long-tailed cache sizes. Both reuse the same single-view frozen features and text anchors as the tables. This comparison transfers the published readout under controlled feature access; the original augmented-cache extraction and trainable Tip-Adapter-F are distinct settings.

On each original base-only pseudo episode, we search $\beta \in \{0.1, 1, 2, 5, 10, 20, 50\}$ and $\alpha \in \{0.1, 1, 5, 10, 20, 50, 100, 200\}$. The native form selects both parameters with zero offset. The calibrated form jointly selects these parameters and an offset from $[-40, 40]$ in steps of 2.5, using macro harmonic accuracy. This 33-point offset grid reflects the raw 100-scaled CLIP score system. No actual novel query or test label is used in selection. Both forms are reported because their selected cache settings can differ. All matched-retention targets are the original retained readout's partition-specific absolute targets.

Table S11: Complete five-shot fixed-gate readout comparison. Values are percentages, mean ± sample standard deviation over three class partitions after averaging the two arrival orders. The paired gain is retained minus comparator average incremental accuracy; a negative gain favors the comparator. Native and calibrated cache forms are selected separately.

Data	Readout	Incremental avg.	Final	Paired avg. gain
CIFAR-100	Retained classifier	82.02 ± 0.84	79.31 ± 1.14	0.00 ± 0.00
CIFAR-100	Classifier removed (fixed gates)	71.40 ± 14.78	68.03 ± 15.63	10.62 ± 15.22
CIFAR-100	No classifier (recalibrated)	79.84 ± 0.82	77.56 ± 1.02	2.18 ± 0.24
CIFAR-100	Anchored text + prototypes	81.56 ± 0.52	79.30 ± 0.59	0.46 ± 0.44
CIFAR-100	Tip-Adapter sum + offset	79.57 ± 0.41	77.21 ± 0.38	2.45 ± 0.63
CIFAR-100	Tip-Adapter mean + offset	81.74 ± 0.93	79.40 ± 1.11	0.28 ± 1.07
CIFAR-100	FeNeC-M + offset	66.60 ± 1.34	63.65 ± 1.55	15.42 ± 0.49
CIFAR-100	FeNeC-E + offset	72.63 ± 1.34	69.00 ± 1.97	9.39 ± 0.68
CIFAR-100	FeNeC-M (native)	59.11 ± 0.85	53.58 ± 1.27	22.91 ± 0.31
CIFAR-100	FeNeC-E (native)	72.63 ± 1.34	69.00 ± 1.97	9.39 ± 0.68
CIFAR-100	Tip-Adapter sum (native)	79.57 ± 0.41	77.21 ± 0.38	2.45 ± 0.63
CIFAR-100	Tip-Adapter mean (native)	81.74 ± 0.93	79.40 ± 1.11	0.28 ± 1.07
Pets	Retained classifier	94.09 ± 2.37	91.77 ± 3.16	0.00 ± 0.00
Pets	Classifier removed (fixed gates)	77.17 ± 26.01	76.85 ± 23.25	16.92 ± 28.33
Pets	No classifier (recalibrated)	94.20 ± 2.05	92.57 ± 2.31	-0.11 ± 0.56
Pets	Anchored text + prototypes	94.35 ± 0.60	92.48 ± 1.10	-0.26 ± 1.82
Pets	Tip-Adapter sum + offset	93.38 ± 2.32	91.50 ± 2.78	0.71 ± 1.21
Pets	Tip-Adapter mean + offset	94.47 ± 2.38	92.88 ± 2.51	-0.37 ± 1.61
Pets	FeNeC-M + offset	80.34 ± 1.29	75.66 ± 1.72	13.76 ± 1.61
Pets	FeNeC-E + offset	84.43 ± 2.37	79.82 ± 2.35	9.67 ± 1.50
Pets	FeNeC-M (native)	73.66 ± 4.73	66.74 ± 6.13	20.44 ± 2.69
Pets	FeNeC-E (native)	84.43 ± 2.37	79.82 ± 2.35	9.67 ± 1.50
Pets	Tip-Adapter sum (native)	94.35 ± 1.08	92.72 ± 1.65	-0.25 ± 1.32
Pets	Tip-Adapter mean (native)	95.15 ± 1.28	93.74 ± 1.29	-1.06 ± 1.33
ImageNet-R	Retained classifier	87.23 ± 0.21	85.40 ± 0.36	0.00 ± 0.00
ImageNet-R	Classifier removed (fixed gates)	77.13 ± 6.75	74.64 ± 7.19	10.09 ± 6.55
ImageNet-R	No classifier (recalibrated)	85.01 ± 0.57	83.65 ± 0.70	2.22 ± 0.40
ImageNet-R	Anchored text + prototypes	86.36 ± 0.24	84.67 ± 0.39	0.87 ± 0.29
ImageNet-R	Tip-Adapter sum + offset	84.62 ± 0.08	82.94 ± 0.07	2.61 ± 0.21
ImageNet-R	Tip-Adapter mean + offset	85.75 ± 0.30	84.19 ± 0.35	1.48 ± 0.35
ImageNet-R	FeNeC-M + offset	59.12 ± 0.54	55.79 ± 0.63	28.11 ± 0.46
ImageNet-R	FeNeC-E + offset	69.66 ± 0.97	66.09 ± 0.62	17.56 ± 0.80
ImageNet-R	FeNeC-M (native)	54.57 ± 0.47	49.71 ± 0.32	32.65 ± 0.37
ImageNet-R	FeNeC-E (native)	69.66 ± 0.97	66.09 ± 0.62	17.56 ± 0.80
ImageNet-R	Tip-Adapter sum (native)	84.62 ± 0.08	82.94 ± 0.07	2.61 ± 0.21
ImageNet-R	Tip-Adapter mean (native)	85.75 ± 0.30	84.19 ± 0.35	1.48 ± 0.35

Appendix G.2. Repeated calibration and support budgets

The 81 configurations span three datasets, three original class partitions, $k \in \{1, 2, 5\}$ and three joint calibration/support trials. Trial zero reuses the original pseudo split. The other trials select $\lfloor |\mathcal{B}|/2 \rfloor$ pseudo-base classes with seeds $1701 + p$ and $2701 + p$. Pseudo classifiers select ridge regularization on pseudo-base validation. Support-index seeds $7, 8, 9$ select one- and two-shot subsets from each five-image reserve. At five shots, the support set is identical across trials: variation comes from the pseudo split. These are correlated trials of the same underlying class partitions. We average the orders and trials within each partition and report the mean and sample standard deviation of the three partition means.

Each configuration independently calibrates the retained, recalibrated-removal, dense-prototype and dynamic-reference readouts on the same base-only episode. Fixed-gate removal preserves the retained mixture's remaining parameters. Dynamic normalization uses the current text candidates and current novel prototypes as references, with the same learned branch and calibration grid as the retained table. Its own mixture is selected independently and stays fixed across arrivals. All retention targets are shared within each configuration. Feasibility counts refer to nine partition/trial combinations per support budget. When only some trials are feasible, the group accuracies average those trials within each partition before summarizing partitions; paired constrained comparisons use the intersection of feasible trial IDs.

Fixed references have zero restricted pre-offset candidate-ranking changes in all 648 append checks (81 configurations, two orders, four appends). The largest floating-point score differences are 1.40×10^{-4} on CIFAR, 5.36×10^{-6} on

Table S12: Calibration and support robustness. Average incremental and final accuracies use fixed gates. Paired gains are retained minus each independently calibrated comparator, after averaging two arrival orders and three trials within each partition. Values are percentages or percentage points, mean ± sample standard deviation across three partitions.

Data	Shots	Retained avg.	Δ recalibrated	Δ dense	Δ dynamic	Retained final
CIFAR-100	1	79.77 ± 0.61	0.55 ± 1.15	0.08 ± 0.72	0.10 ± 0.74	76.32 ± 1.22
CIFAR-100	2	80.20 ± 0.32	0.84 ± 0.67	-0.76 ± 0.52	0.55 ± 0.84	77.00 ± 0.85
CIFAR-100	5	80.57 ± 0.96	1.22 ± 1.18	-0.79 ± 1.08	0.62 ± 0.59	77.25 ± 1.09
Pets	1	93.94 ± 2.37	0.05 ± 0.67	-0.24 ± 0.48	-0.54 ± 0.11	91.60 ± 3.20
Pets	2	92.64 ± 1.82	-1.45 ± 1.19	-1.62 ± 1.15	-1.43 ± 1.09	90.09 ± 1.87
Pets	5	94.56 ± 1.74	0.78 ± 1.03	0.43 ± 0.80	0.47 ± 1.43	92.75 ± 2.00
ImageNet-R	1	86.16 ± 0.22	1.81 ± 0.33	1.01 ± 0.29	-0.07 ± 0.29	83.83 ± 0.14
ImageNet-R	2	86.58 ± 0.37	2.10 ± 0.28	0.62 ± 0.34	-0.07 ± 0.28	84.43 ± 0.67
ImageNet-R	5	87.16 ± 0.12	2.14 ± 0.37	0.54 ± 0.23	0.15 ± 0.19	85.36 ± 0.25

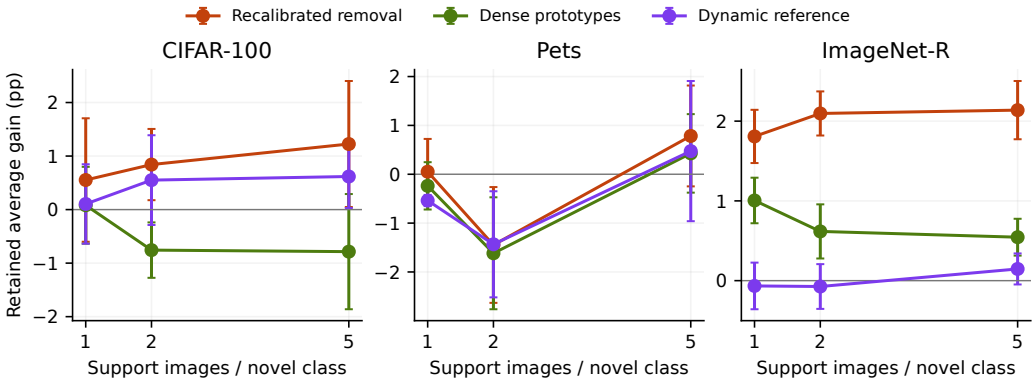


Figure S4: Paired average incremental gains under one-, two- and five-shot support. Error bars are partition-level sample standard deviations after averaging correlated calibration/support trials and arrival orders. The horizontal zero line identifies which readout has higher average accuracy in each setting.

Pets and 2.19×10^{-5} on ImageNet-R. Dynamic references produce maximum score changes of 2.40, 1.78 and 1.30, respectively, and up to 429, 206 and 118 restricted prediction changes per append over the full cached test query pool. These diagnostics compare the same previously active candidate IDs before adding the fixed group offset; they isolate changes induced by reference statistics. Recognition accuracy is reported separately in the fixed-gate and retention-policy tables.

Table S13: CIFAR-100: final-session outcomes in the repeated-calibration study at the 95% budget. Match uses base test labels descriptively; Val. uses base validation only and reports achieved test accuracy. Feasibility is counted over nine correlated partition/trial combinations.

Shots	Readout	Match base	Match novel	Feasible	Val. base	Val. novel	Feasible
1	Retained classifier	83.54 ± 0.81	73.03 ± 3.08	9/9	80.77 ± 2.34	76.30 ± 3.82	9/9
1	No classifier (recalibrated)	83.37 ± 0.57	59.17 ± 3.51	8/9	80.92 ± 1.45	72.28 ± 4.17	9/9
1	Anchored text + prototypes	83.37 ± 0.57	63.65 ± 3.03	8/9	80.38 ± 2.28	73.74 ± 5.16	9/9
1	Dynamic reference	83.54 ± 0.81	73.07 ± 3.81	9/9	81.10 ± 2.54	75.86 ± 4.56	9/9
2	Retained classifier	82.99 ± 0.46	73.84 ± 3.52	9/9	80.65 ± 1.93	76.40 ± 4.33	9/9
2	No classifier (recalibrated)	82.99 ± 0.46	61.20 ± 0.62	9/9	81.14 ± 1.01	71.44 ± 2.35	9/9
2	Anchored text + prototypes	82.99 ± 0.46	71.62 ± 6.74	9/9	80.81 ± 1.65	76.62 ± 4.51	9/9
2	Dynamic reference	82.99 ± 0.46	74.70 ± 3.56	9/9	80.37 ± 1.53	77.66 ± 2.64	9/9
5	Retained classifier	83.83 ± 0.76	75.71 ± 2.77	9/9	82.03 ± 1.38	77.69 ± 1.64	9/9
5	No classifier (recalibrated)	83.55 ± 0.29	62.19 ± 6.29	7/9	81.21 ± 1.18	73.83 ± 6.90	7/9
5	Anchored text + prototypes	83.83 ± 0.76	73.28 ± 3.14	9/9	81.95 ± 0.54	75.45 ± 1.14	8/9
5	Dynamic reference	83.83 ± 0.76	73.83 ± 0.74	9/9	82.44 ± 1.94	76.58 ± 2.18	9/9

Table S14: Oxford-IIIT Pets: final-session outcomes under the same repeated-calibration policies and aggregation as Table S13.

Shots	Readout	Match base	Match novel	Feasible	Val. base	Val. novel	Feasible
1	Retained classifier	92.58 ± 1.83	93.50 ± 0.89	9/9	88.07 ± 1.16	94.84 ± 0.89	9/9
1	No classifier (recalibrated)	92.58 ± 1.85	92.30 ± 2.36	9/9	88.56 ± 1.16	95.45 ± 0.70	6/9
1	Anchored text + prototypes	92.57 ± 1.84	93.03 ± 1.60	9/9	87.59 ± 1.34	94.59 ± 1.55	9/9
1	Dynamic reference	92.56 ± 1.82	93.74 ± 1.02	9/9	87.89 ± 1.50	95.47 ± 0.16	8/9
2	Retained classifier	92.28 ± 2.51	94.42 ± 1.14	9/9	88.05 ± 2.52	95.66 ± 0.49	9/9
2	No classifier (recalibrated)	92.27 ± 2.52	93.97 ± 0.96	9/9	87.04 ± 3.11	95.89 ± 0.27	7/9
2	Anchored text + prototypes	92.27 ± 2.50	94.33 ± 0.58	9/9	87.64 ± 2.55	95.51 ± 0.58	9/9
2	Dynamic reference	92.27 ± 2.50	94.62 ± 1.31	9/9	87.85 ± 2.40	95.77 ± 0.27	9/9
5	Retained classifier	92.40 ± 2.41	94.95 ± 1.39	9/9	88.31 ± 2.40	96.06 ± 0.47	9/9
5	No classifier (recalibrated)	92.40 ± 2.42	94.08 ± 0.92	9/9	87.39 ± 2.71	96.12 ± 0.41	7/9
5	Anchored text + prototypes	92.39 ± 2.39	94.79 ± 0.92	9/9	87.94 ± 2.68	95.99 ± 0.80	9/9
5	Dynamic reference	92.40 ± 2.40	94.84 ± 1.36	9/9	87.44 ± 2.34	96.12 ± 0.27	9/9

Table S15: ImageNet-R: final-session outcomes under the same repeated-calibration policies and aggregation as Table S13.

Shots	Readout	Match base	Match novel	Feasible	Val. base	Val. novel	Feasible
1	Retained classifier	86.24 ± 0.38	81.59 ± 0.50	9/9	85.73 ± 0.71	81.95 ± 0.67	9/9
1	No classifier (recalibrated)	86.25 ± 0.37	70.69 ± 10.24	9/9	86.13 ± 0.37	69.56 ± 13.12	9/9
1	Anchored text + prototypes	86.24 ± 0.36	78.57 ± 0.59	9/9	86.14 ± 0.54	78.97 ± 0.47	9/9
1	Dynamic reference	86.25 ± 0.36	81.78 ± 0.76	9/9	85.87 ± 0.61	82.11 ± 1.07	9/9
2	Retained classifier	86.38 ± 0.34	82.67 ± 1.21	9/9	86.10 ± 0.63	82.86 ± 1.30	9/9
2	No classifier (recalibrated)	86.38 ± 0.33	69.12 ± 14.06	9/9	86.27 ± 0.38	58.73 ± 33.47	9/9
2	Anchored text + prototypes	86.39 ± 0.34	81.53 ± 2.15	9/9	86.40 ± 0.40	81.50 ± 1.68	9/9
2	Dynamic reference	86.38 ± 0.33	82.80 ± 0.65	9/9	86.04 ± 0.67	83.02 ± 0.76	9/9
5	Retained classifier	86.35 ± 0.27	84.48 ± 0.35	9/9	86.08 ± 0.44	84.69 ± 0.46	9/9
5	No classifier (recalibrated)	86.34 ± 0.27	73.89 ± 9.72	9/9	86.36 ± 0.21	63.98 ± 26.37	9/9
5	Anchored text + prototypes	86.34 ± 0.27	83.45 ± 1.62	9/9	86.23 ± 0.52	83.65 ± 1.19	9/9
5	Dynamic reference	86.35 ± 0.29	84.42 ± 0.56	9/9	86.21 ± 0.47	84.50 ± 0.69	9/9

Appendix G.3. Readout cost and restart verification

Table S16 measures 512 cached queries on the RTX 4070 Laptop in float32. Each readout receives five warm-ups and 20 synchronized timing repetitions. Static cache values are constructed before timing, query chunks contain 256 images for retrieval, and unused score branches are skipped. An independently checked implementation verifies score agreement within 10^{-3} for every optimized readout. Persistent tensor bytes count text anchors, learned weights and references/prototypes for the tables, or text anchors, cache keys and the stored dense one-hot/class-mean value matrix for retrieval. These are implementation-level readout costs, excluding the common image encoder, query features and temporary workspaces. They do not represent complete model memory or end-to-end image inference time.

Every calibration and arrival-session result is written atomically. A live pause/resume check preserved a completed session's SHA-256 hash and modification time, then completed all 90 configuration units. The numerical verifier reconstructs group predictions, macro weights, selected thresholds and test metrics from archived sufficient statistics, including the base-validation

Table S16: Measured readout latency and persistent tensor storage for a final-session 512-query batch on the same laptop GPU. MiB denotes 2^{20} bytes. Cache keys number 4,937 on CIFAR, 553 on Pets and 3,830 on ImageNet-R in partition zero.

Data	Readout	Median (ms)	P90 (ms)	Stored tensors (MiB)
CIFAR-100	Retained classifier	0.715	0.763	0.586
CIFAR-100	Dense text + prototypes	0.515	0.542	0.469
CIFAR-100	Tip-Adapter sum	0.838	0.862	16.205
CIFAR-100	Tip-Adapter mean	0.841	0.857	16.205
Pets	Retained classifier	0.880	1.280	0.220
Pets	Dense text + prototypes	0.586	0.680	0.176
Pets	Tip-Adapter sum	0.564	0.673	1.771
Pets	Tip-Adapter mean	0.569	0.732	1.771
ImageNet-R	Retained classifier	0.864	0.918	1.465
ImageNet-R	Dense text + prototypes	0.631	0.790	1.172
ImageNet-R	Tip-Adapter sum	0.759	0.873	14.729
ImageNet-R	Tip-Adapter mean	0.732	0.898	14.729

selection arrays. The trial-zero five-shot table results agree exactly with the prior three-dataset records. Source, protocol and cache hashes, all outcomes and reproduction commands accompany the release.

Appendix H. Validation-query budget for admission thresholds

This experiment varies the old-class query labels used to select the admission threshold, conditional on the original five-shot classifier and fusion calibration. All fitted parameters, cache settings and branch mixtures stay fixed. The evaluation uses the original trial-zero score archives, class partitions and arrival order zero on each dataset. No image encoder or new classifier is fitted.

For each base class with n_c validation images, the nominal fraction f keeps $\max(1, \lfloor f n_c \rfloor)$ images. Fractions $f \in \{0.25, 0.5\}$ use three predeclared subset seeds, with nested class-wise permutations; the full budget uses all images once. Actual query counts are reported because rounding and the one-query

minimum change the realized fractions. Macro weights are recomputed from the selected class counts. The retained readout's closed-base accuracy on that subset defines a shared target $rA^{\text{closed}}_{B,\text{ref,subset}}$ for all six readouts, with $r \in \{0.95, 0.99\}$. The largest feasible offset uses only the selected base-validation margins and labels.

After selection, test queries measure both group accuracies. A test target is met when $A_{B,\text{test}} \geq rA^{\text{closed}}_{B,\text{ref,test}}$; this criterion is evaluated after the offset is fixed and never enters threshold selection. Table S17 separates validation feasibility from test target attainment. Subset repeats are averaged within each class partition before the mean and sample standard deviation over available partition means. The nine reduced-budget cases share three underlying partitions and are not independent repetitions. Small per-class counts quantize the selected threshold, so different target ratios can yield identical predictions and increasing the validation fraction need not change test target attainment monotonically. These empirical policies do not supply a finite-sample guarantee on test retention.

The full-budget policies reproduce the original validation-selected outcomes exactly. Saved predictions reconstruct every offset, macro accuracy and attainment flag across 315 session units and 3,780 metric records. The complete budgets, query fractions and test-retention ratios are in `validation_budget_summary.csv`. This experiment varies threshold-selection queries while holding the initial full-data fusion calibration fixed.

Table S17: Retained readout under different validation-query budgets after the final arrival. Accuracies are percentages, mean ± partition-level sample standard deviation after averaging subset repeats. Queries gives the actual old-class validation count range. Val. feasible counts partition/subset combinations meeting the validation constraint; Test met counts feasible combinations meeting the separate test criterion. The full budget uses one subset per partition.

Data	Nominal budget	Queries	Target	Base test	Novel test	Val. feasible	Test met
CIFAR-100	25%	40	95%	82.46 ± 2.25	75.14 ± 3.50	9/9	4/9
CIFAR-100	25%	40	99%	85.21 ± 2.64	66.99 ± 9.18	9/9	4/9
CIFAR-100	50%	68	95%	81.94 ± 3.23	76.92 ± 3.72	9/9	3/9
CIFAR-100	50%	68	99%	85.85 ± 2.61	64.30 ± 10.71	9/9	5/9
CIFAR-100	100%	161–163	95%	82.99 ± 2.29	75.71 ± 3.14	3/3	1/3
CIFAR-100	100%	161–163	99%	86.48 ± 1.65	62.83 ± 10.11	3/3	2/3
Pets	25%	15	95%	91.77 ± 2.39	92.42 ± 2.64	9/9	5/9
Pets	25%	15	99%	91.77 ± 2.39	92.42 ± 2.64	9/9	4/9
Pets	50%	25	95%	92.02 ± 0.95	94.75 ± 1.23	9/9	5/9
Pets	50%	25	99%	95.05 ± 0.48	90.49 ± 5.09	9/9	5/9
Pets	100%	60	95%	90.29 ± 2.90	95.68 ± 0.83	3/3	0/3
Pets	100%	60	99%	96.43 ± 2.15	88.67 ± 7.89	3/3	2/3
ImageNet-R	25%	696–731	95%	85.85 ± 0.38	84.98 ± 0.38	9/9	2/9
ImageNet-R	25%	696–731	99%	89.79 ± 0.16	74.81 ± 2.33	9/9	5/9
ImageNet-R	50%	1444–1519	95%	85.93 ± 0.65	84.87 ± 0.63	9/9	1/9
ImageNet-R	50%	1444–1519	99%	89.90 ± 0.33	74.24 ± 2.68	9/9	5/9
ImageNet-R	100%	2938–3087	95%	86.01 ± 0.59	84.73 ± 0.71	3/3	0/3
ImageNet-R	100%	2938–3087	99%	89.94 ± 0.25	74.48 ± 1.35	3/3	2/3

Table S18: All readouts at the nominal 25% validation-query budget and 95% target after the final arrival. All classifiers, mixtures and cache settings are held fixed from the original five-shot calibration. Accuracy aggregation and feasibility/attainment counts follow Table S17. Missing metrics correspond to zero feasible subsets.

Data	Readout	Base test	Novel test	Val. feasible	Test met
CIFAR-100	Retained classifier	82.46 ± 2.25	75.14 ± 3.50	9/9	4/9
CIFAR-100	Fixed-gate removal	79.39 ± 2.08	74.84 ± 0.15	5/9	1/5
CIFAR-100	Recalibrated removal	79.48 ± 2.19	73.22 ± 2.45	5/9	1/5
CIFAR-100	Dense text + prototypes	81.30 ± 2.08	77.12 ± 2.64	7/9	0/7
CIFAR-100	Tip-Adapter sum	79.32 ± 2.30	71.40 ± 7.86	6/9	1/6
CIFAR-100	Tip-Adapter mean	79.51 ± 3.58	77.12 ± 5.00	8/9	1/8
Pets	Retained classifier	91.77 ± 2.39	92.42 ± 2.64	9/9	5/9
Pets	Fixed-gate removal	88.87 ± 3.95	92.77 ± 5.91	6/9	1/6
Pets	Recalibrated removal	87.68 ± 3.95	89.64 ± 11.35	6/9	1/6
Pets	Dense text + prototypes	88.50 ± 6.27	93.41 ± 3.82	7/9	2/7
Pets	Tip-Adapter sum	89.86 ± 3.58	89.59 ± 5.68	7/9	2/7
Pets	Tip-Adapter mean	90.25 ± 1.80	91.90 ± 3.31	7/9	2/7
ImageNet-R	Retained classifier	85.85 ± 0.38	84.98 ± 0.38	9/9	2/9
ImageNet-R	Fixed-gate removal	85.66 ± 0.86	79.23 ± 1.90	7/9	3/7
ImageNet-R	Recalibrated removal	85.74 ± 0.90	79.08 ± 1.74	7/9	3/7
ImageNet-R	Dense text + prototypes	86.39 ± 0.50	83.05 ± 0.22	9/9	4/9
ImageNet-R	Tip-Adapter sum	85.47 ± 0.89	77.71 ± 1.44	8/9	4/8
ImageNet-R	Tip-Adapter mean	86.49 ± 0.54	81.22 ± 0.63	9/9	6/9

Appendix I. Ten-partition replication, encoder transfer and simple fusion

The expanded experiment fixes class-partition seeds $0, \ldots, 9$ on each of the three existing image pools. Assignments and pseudo splits $0, 1, 2$ are retained exactly, and the original L/14 results reconstruct with maximum metric difference zero. New seeds use the same frequency-stratified allocation, base/novel counts, five-shot reserve, pseudo-episode selection and two arrival-order rules. These are ten distinct class assignments on shared images, not ten independent datasets. Primary fixed-gate comparisons average both orders within a partition. The conditional paired percentile bootstrap uses 20,000 resamples of the ten paired partition values with seed 2026. Constrained differences use identical jointly feasible partition/order/session cells. JSON records retain the partition values and available sets, and CSV tables report every budget.

Appendix I.1. Second encoder and access controls

OpenAI CLIP ViT-B/32 uses cached revision `3d74acf9a28c67741b2f4f2ea7635f0aaf6f0268` with 512-dimensional projected features; L/14 uses 768 dimensions. Both consume the exact same fitting, validation, support and test images, dataset-specific preprocessing and eight text templates. Text anchors and ridge columns are recomputed for the selected encoder; no task-adapted features are used. The two encoders have separate closed-base references and hence separate absolute targets. Config, weights, tokenizer and pool hashes are recorded. Batch-16 feature shards support atomic restart with contiguous-index guards, and eight fresh test images are re-encoded per dataset to check each completed B/32 cache.

Appendix I.2. Raw and fixed-temperature controls

Raw fusion uses q_T, q_L and q_P directly in the same availability pattern as the retained table. Fixed-temperature fusion uses q_e/σ_e, where

$$\sigma_e = \max\left(10^{-6}, \frac{1}{|\mathcal{V}_B|}\sum_{x\in\mathcal{V}_B} \mathrm{Std}_{r\in\mathcal{R}_e}\, q_e(x,r)\right).$$

For the pseudo episode, $\mathcal{V}_B$ contains base-validation queries and $\mathcal{R}_e$ contains pseudo-base reference columns. Full-model scales use full base columns and base validation before arrivals. The three scalars remain fixed across sessions. Coefficient and offset grids are identical to the main retained model, with the same harmonic pseudo-episode objective. Neither control requires current-vocabulary statistics; both can preserve old scores. Their performance tests the empirical value of the per-query normalization, conditional on this specified calibration grid.

Appendix I.3. Full matched and validation-policy results

The main validation table includes retained, recalibrated-removal, dense-prototype, class-mean cache and simple-fusion alternatives. Summed cache and fixed-gate removal remain in the complete records. The following tables give both encoders' matched-test frontiers and the B/32 validation policy at the 95% budget. Every feasible set and test-attainment count refers to the final active vocabulary; order zero counts each partition once. Because means may cover different feasible sets, an unpaired difference of available means is not reported as a paired gain. Complete 80/85/90/95/99% outcomes are retained in `revision_summary.csv`; per-partition raw predictions supply both test and validation sufficient statistics.

Table S19: Complete eight-readout ViT-L/14 validation-selected 95% policy. Accuracies are achieved test percentages, mean ± sample standard deviation over feasible partitions. Feasible and Test met have the same meanings as the main validation-policy table.

Data	Readout	Base	Novel	Feasible	Test met
CIFAR-100	Retained references	84.26 ± 1.91	77.10 ± 2.53	10/10	7/10
CIFAR-100	Fixed-gate removal	83.05 ± 1.64	65.32 ± 12.41	6/10	1/6
CIFAR-100	Recalibrated removal	82.91 ± 1.62	66.03 ± 13.04	6/10	1/6
CIFAR-100	Dense text + prototypes	84.00 ± 1.76	75.10 ± 7.15	8/10	4/8
CIFAR-100	Raw-score fusion	82.11 ± 2.51	64.45 ± 13.35	7/10	1/7
CIFAR-100	Fixed temperatures	83.72 ± 2.24	70.93 ± 10.56	10/10	5/10
CIFAR-100	Tip-Adapter sum	83.21 ± 1.71	65.36 ± 12.68	5/10	0/5
CIFAR-100	Tip-Adapter mean	83.19 ± 1.72	68.69 ± 10.39	9/10	2/9
Pets	Retained references	89.95 ± 2.67	93.79 ± 3.83	10/10	1/10
Pets	Fixed-gate removal	90.43 ± 3.16	93.81 ± 3.11	8/10	3/8
Pets	Recalibrated removal	90.48 ± 3.04	93.44 ± 3.56	8/10	3/8
Pets	Dense text + prototypes	89.46 ± 3.08	94.47 ± 2.60	10/10	2/10
Pets	Raw-score fusion	91.35 ± 2.91	92.72 ± 2.70	9/10	4/9
Pets	Fixed temperatures	90.48 ± 3.42	94.21 ± 3.19	9/10	3/9
Pets	Tip-Adapter sum	89.18 ± 1.73	91.61 ± 6.00	7/10	1/7
Pets	Tip-Adapter mean	90.73 ± 2.17	91.39 ± 4.32	10/10	4/10
ImageNet-R	Retained references	86.15 ± 0.94	84.63 ± 0.95	10/10	2/10
ImageNet-R	Fixed-gate removal	86.17 ± 0.65	66.89 ± 22.80	10/10	3/10
ImageNet-R	Recalibrated removal	86.16 ± 0.71	66.88 ± 22.79	10/10	3/10
ImageNet-R	Dense text + prototypes	86.01 ± 0.94	83.94 ± 1.10	10/10	3/10
ImageNet-R	Raw-score fusion	86.15 ± 0.68	65.55 ± 20.42	10/10	2/10
ImageNet-R	Fixed temperatures	85.99 ± 0.71	82.92 ± 1.03	10/10	1/10
ImageNet-R	Tip-Adapter sum	86.06 ± 0.78	74.52 ± 3.69	10/10	2/10
ImageNet-R	Tip-Adapter mean	86.08 ± 1.08	81.61 ± 1.51	10/10	4/10

Table S20: ViT-L/14 matched-test 95% retention frontier across ten class partitions. Values are percentages, mean ± sample standard deviation over feasible partitions after averaging orders. Test met equals feasibility by construction in this descriptive sweep.

Data	Readout	Base	Novel	Feasible	Test met
CIFAR-100	Retained references	83.91 ± 1.42	78.02 ± 2.27	10/10	10/10
CIFAR-100	Fixed-gate removal	83.25 ± 1.17	61.68 ± 9.35	6/10	6/6
CIFAR-100	Recalibrated removal	83.25 ± 1.17	63.78 ± 6.52	6/10	6/6
CIFAR-100	Dense text + prototypes	83.91 ± 1.42	76.18 ± 2.56	10/10	10/10
CIFAR-100	Raw-score fusion	83.25 ± 1.17	62.37 ± 6.38	6/10	6/6
CIFAR-100	Fixed temperatures	83.64 ± 1.21	74.53 ± 2.70	9/10	9/9
CIFAR-100	Tip-Adapter sum	83.16 ± 1.28	61.83 ± 4.92	5/10	5/5
CIFAR-100	Tip-Adapter mean	83.91 ± 1.42	65.20 ± 11.69	10/10	10/10
Pets	Retained references	91.56 ± 1.87	93.30 ± 3.74	10/10	10/10
Pets	Fixed-gate removal	91.57 ± 1.88	93.78 ± 2.17	10/10	10/10
Pets	Recalibrated removal	91.56 ± 1.88	93.32 ± 2.10	10/10	10/10
Pets	Dense text + prototypes	91.56 ± 1.87	93.79 ± 2.78	10/10	10/10
Pets	Raw-score fusion	91.57 ± 1.88	92.39 ± 2.75	10/10	10/10
Pets	Fixed temperatures	91.56 ± 1.87	93.64 ± 2.48	10/10	10/10
Pets	Tip-Adapter sum	91.51 ± 1.98	87.78 ± 9.98	9/10	9/9
Pets	Tip-Adapter mean	91.57 ± 1.87	92.02 ± 4.15	10/10	10/10
ImageNet-R	Retained references	86.42 ± 0.53	84.31 ± 1.01	10/10	10/10
ImageNet-R	Fixed-gate removal	86.41 ± 0.56	73.82 ± 6.70	9/10	9/9
ImageNet-R	Recalibrated removal	86.42 ± 0.56	73.50 ± 6.79	9/10	9/9
ImageNet-R	Dense text + prototypes	86.42 ± 0.53	83.02 ± 1.40	10/10	10/10
ImageNet-R	Raw-score fusion	86.42 ± 0.56	70.85 ± 8.03	9/10	9/9
ImageNet-R	Fixed temperatures	86.43 ± 0.53	82.49 ± 1.11	10/10	10/10
ImageNet-R	Tip-Adapter sum	86.43 ± 0.56	73.16 ± 5.66	9/10	9/9
ImageNet-R	Tip-Adapter mean	86.43 ± 0.52	80.91 ± 1.48	10/10	10/10

Table S21: ViT-B/32 matched-test 95% frontier under the same access, aggregation and feasibility reporting as Table S20. Absolute targets are encoder-specific.

Data	Readout	Base	Novel	Feasible	Test met
CIFAR-100	Retained references	76.16 ± 1.99	59.64 ± 2.10	10/10	10/10
CIFAR-100	Fixed-gate removal	74.84 ± 1.75	49.81 ± 10.48	4/10	4/4
CIFAR-100	Recalibrated removal	74.84 ± 1.75	49.81 ± 10.48	4/10	4/4
CIFAR-100	Dense text + prototypes	76.15 ± 2.22	51.78 ± 8.42	8/10	8/8
CIFAR-100	Raw-score fusion	75.12 ± 1.63	46.30 ± 9.93	5/10	5/5
CIFAR-100	Fixed temperatures	75.71 ± 1.46	54.14 ± 4.58	9/10	9/9
CIFAR-100	Tip-Adapter sum	74.08 ± 1.05	52.81 ± 5.29	3/10	3/3
CIFAR-100	Tip-Adapter mean	76.16 ± 2.11	46.50 ± 11.77	9/10	9/9
Pets	Retained references	86.92 ± 2.17	76.54 ± 6.71	10/10	10/10
Pets	Fixed-gate removal	86.45 ± 2.27	77.62 ± 6.10	5/10	5/5
Pets	Recalibrated removal	86.46 ± 2.29	78.61 ± 5.07	5/10	5/5
Pets	Dense text + prototypes	86.84 ± 2.33	71.48 ± 15.80	8/10	8/8
Pets	Raw-score fusion	87.12 ± 2.19	67.25 ± 11.41	9/10	9/9
Pets	Fixed temperatures	86.93 ± 2.16	74.97 ± 6.20	10/10	10/10
Pets	Tip-Adapter sum	86.83 ± 2.22	70.08 ± 9.72	6/10	6/6
Pets	Tip-Adapter mean	87.28 ± 2.29	68.64 ± 21.10	8/10	8/8
ImageNet-R	Retained references	70.38 ± 0.87	56.36 ± 1.11	10/10	10/10
ImageNet-R	Fixed-gate removal	69.72	31.36	1/10	1/1
ImageNet-R	Recalibrated removal	69.71	30.54	1/10	1/1
ImageNet-R	Dense text + prototypes	70.37 ± 0.88	48.30 ± 3.45	10/10	10/10
ImageNet-R	Raw-score fusion	69.71	28.06	1/10	1/1
ImageNet-R	Fixed temperatures	70.38 ± 0.87	53.72 ± 1.35	10/10	10/10
ImageNet-R	Tip-Adapter sum	69.67 ± 0.03	35.63 ± 8.58	2/10	2/2
ImageNet-R	Tip-Adapter mean	70.45 ± 0.92	41.88 ± 7.58	8/10	8/8

Table S22: ViT-B/32 validation-selected 95% policy. Values are achieved base/novel test percentages over validation-feasible partitions; Test met counts those meeting the independent test-retention criterion. Validation feasibility does not guarantee test target attainment.

Data	Readout	Base	Novel	Feasible	Test met
CIFAR-100	Retained references	75.98 ± 2.43	59.04 ± 3.66	10/10	5/10
CIFAR-100	Fixed-gate removal	73.46 ± 1.73	52.54 ± 10.71	7/10	1/7
CIFAR-100	Recalibrated removal	73.40 ± 1.79	54.20 ± 6.94	7/10	1/7
CIFAR-100	Dense text + prototypes	74.30 ± 2.85	55.73 ± 8.03	7/10	2/7
CIFAR-100	Raw-score fusion	73.82 ± 2.67	49.47 ± 12.37	8/10	2/8
CIFAR-100	Fixed temperatures	73.90 ± 2.39	56.16 ± 8.19	9/10	1/9
CIFAR-100	Tip-Adapter sum	73.55 ± 2.53	49.25 ± 12.78	7/10	1/7
CIFAR-100	Tip-Adapter mean	73.54 ± 2.88	57.47 ± 8.46	7/10	0/7
Pets	Retained references	88.16 ± 3.34	72.75 ± 8.65	10/10	8/10
Pets	Fixed-gate removal	86.17 ± 2.74	76.30 ± 6.46	5/10	3/5
Pets	Recalibrated removal	86.24 ± 2.14	77.70 ± 5.37	5/10	2/5
Pets	Dense text + prototypes	86.89 ± 3.09	70.27 ± 15.78	7/10	3/7
Pets	Raw-score fusion	86.93 ± 5.36	63.79 ± 15.01	9/10	5/9
Pets	Fixed temperatures	85.53 ± 5.27	74.57 ± 7.40	10/10	4/10
Pets	Tip-Adapter sum	86.87 ± 2.42	73.74 ± 4.16	4/10	2/4
Pets	Tip-Adapter mean	84.97 ± 3.13	81.94 ± 3.51	5/10	1/5
ImageNet-R	Retained references	70.18 ± 0.96	56.80 ± 1.33	10/10	3/10
ImageNet-R	Fixed-gate removal	68.23 ± 0.84	19.12 ± 6.82	2/10	0/2
ImageNet-R	Recalibrated removal	68.23 ± 0.84	19.12 ± 6.82	2/10	0/2
ImageNet-R	Dense text + prototypes	70.43 ± 1.50	46.95 ± 4.97	10/10	6/10
ImageNet-R	Raw-score fusion	68.16 ± 0.84	21.79 ± 6.87	2/10	0/2
ImageNet-R	Fixed temperatures	70.46 ± 1.26	53.38 ± 1.43	10/10	6/10
ImageNet-R	Tip-Adapter sum	68.89 ± 0.79	38.12 ± 2.27	3/10	0/3
ImageNet-R	Tip-Adapter mean	69.99 ± 1.61	46.39 ± 3.62	9/10	4/9

Table S23: Full ten-partition paired average and final gains. Entries are retained-minus-comparator percentage points [conditional paired 95% bootstrap CI], with orders averaged within each partition.

Encoder	Data	Comparator	Paired avg. gain [95% CI]	Paired final gain [95% CI]
L/14	CIFAR-100	Recalibrated removal	3.00 [2.23, 3.75]	2.62 [1.81, 3.43]
L/14	CIFAR-100	Dense text + prototypes	0.15 [-0.46, 0.64]	-0.21 [-0.85, 0.30]
L/14	CIFAR-100	Tip-Adapter mean	1.44 [0.68, 2.10]	1.24 [0.41, 2.01]
L/14	Pets	Recalibrated removal	-0.14 [-1.53, 0.86]	-0.78 [-2.84, 0.60]
L/14	Pets	Dense text + prototypes	-0.98 [-2.52, 0.22]	-1.54 [-4.08, 0.35]
L/14	Pets	Tip-Adapter mean	-0.06 [-1.76, 1.41]	-0.39 [-2.92, 1.71]
L/14	ImageNet-R	Recalibrated removal	2.30 [1.95, 2.59]	1.71 [1.28, 2.06]
L/14	ImageNet-R	Dense text + prototypes	0.71 [0.46, 0.92]	0.41 [0.03, 0.72]
L/14	ImageNet-R	Tip-Adapter mean	1.42 [1.15, 1.62]	1.13 [0.76, 1.43]
B/32	CIFAR-100	Recalibrated removal	2.38 [1.44, 3.34]	1.80 [1.01, 2.65]
B/32	CIFAR-100	Dense text + prototypes	0.86 [-0.15, 1.76]	0.43 [-0.47, 1.24]
B/32	CIFAR-100	Tip-Adapter mean	0.96 [-0.04, 1.84]	0.51 [-0.34, 1.33]
B/32	Pets	Recalibrated removal	-0.44 [-1.64, 0.81]	-2.00 [-3.80, -0.19]
B/32	Pets	Dense text + prototypes	-0.39 [-1.85, 1.02]	-1.17 [-3.22, 0.67]
B/32	Pets	Tip-Adapter mean	-0.74 [-1.62, 0.12]	-1.85 [-3.40, -0.59]
B/32	ImageNet-R	Recalibrated removal	3.07 [2.82, 3.34]	2.39 [2.19, 2.61]
B/32	ImageNet-R	Dense text + prototypes	0.90 [0.69, 1.11]	0.46 [0.26, 0.66]
B/32	ImageNet-R	Tip-Adapter mean	1.71 [1.44, 1.98]	1.34 [1.16, 1.52]

Table S24: Complete L/14 normalization controls. Fixed-gate averages cover five sessions; matched base/novel accuracy uses the descriptive 95% test-retention budget. Means and sample standard deviations use the reported feasible sets.

Data	Fusion	Fixed-gate avg.	Matched base	Matched novel	Feasible
CIFAR-100	Retained references	83.03 ± 1.19	83.91 ± 1.42	78.02 ± 2.27	10/10
CIFAR-100	Raw-score fusion	79.70 ± 2.64	83.25 ± 1.17	62.37 ± 6.38	6/10
CIFAR-100	Fixed temperatures	81.56 ± 0.98	83.64 ± 1.21	74.53 ± 2.70	9/10
Pets	Retained references	92.23 ± 2.87	91.56 ± 1.87	93.30 ± 3.74	10/10
Pets	Raw-score fusion	87.94 ± 14.58	91.57 ± 1.88	92.39 ± 2.75	10/10
Pets	Fixed temperatures	89.39 ± 10.15	91.56 ± 1.87	93.64 ± 2.48	10/10
ImageNet-R	Retained references	87.24 ± 0.59	86.42 ± 0.53	84.31 ± 1.01	10/10
ImageNet-R	Raw-score fusion	83.70 ± 2.58	86.42 ± 0.56	70.85 ± 8.03	9/10
ImageNet-R	Fixed temperatures	86.30 ± 0.38	86.43 ± 0.53	82.49 ± 1.11	10/10

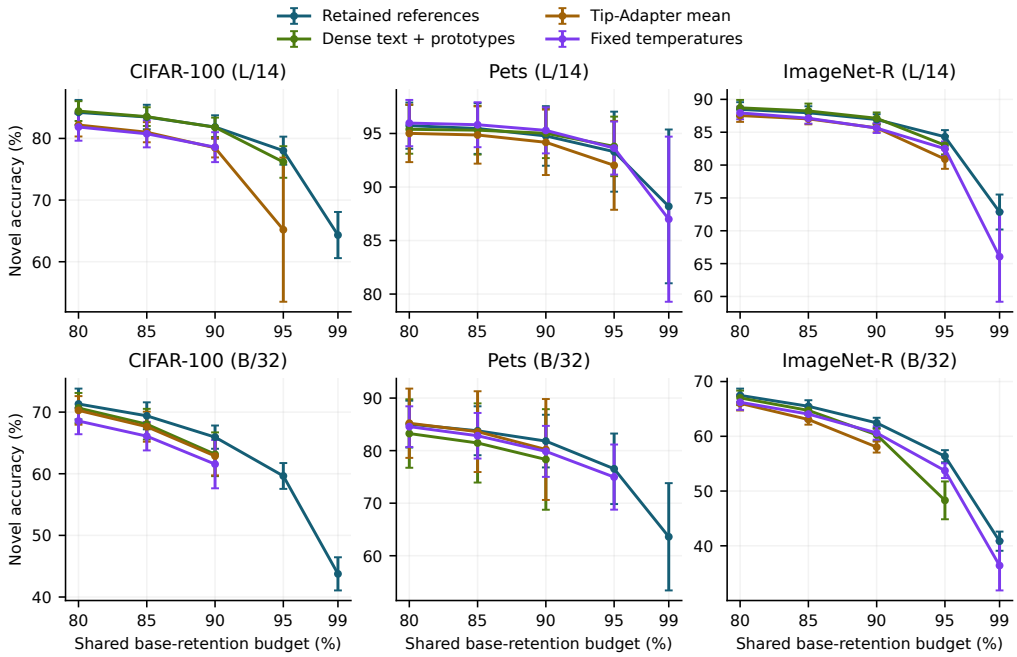


Figure S5: Descriptive matched-retention capacity. Readouts share the retained model's absolute base target within each partition/encoder. Error bars show partition standard deviations; plotted points require all ten partitions to be feasible. Offsets use test labels for this capacity analysis; tables in this section include partially feasible sets.

Appendix J. Independent protection-risk calibration

Appendix J.1. Disjoint access and refitting

For each dataset, original fitting and validation features are concatenated in that order. A generator with seed 662026 shuffles each class, reserving $\max(1, \lfloor 0.4N_c \rfloor)$ calibration and $\max(1, \lfloor 0.2N_c \rfloor)$ tuning observations; the remainder fits the readouts. Every class retains at least two fitting images. Identical indices are used by both encoders and every method. The global fitting/tuning/calibration sizes are 4,437/2,139/4,318 on CIFAR, 556/248/510 on Pets, and 5,096/2,390/4,863 on ImageNet-R. Each true partition uses only

its base classes in these pools. Support and test observations remain outside all three. All ridge parameters, pseudo-episode mixtures, branch temperatures and semantic-cache hyperparameters are recomputed using fitting and tuning data exclusively.

Only frozen features are reused; previously tuned score models are not treated as independent of their old validation labels. The new calibration labels enter only threshold certification or the explicitly compared empirical threshold controls. The independent-pool experiment is a benchmark revision on previously studied test data, not a newly untouched final evaluation. The statistical analysis assumes that, conditional on each base class and the fitted models, calibration observations are independent draws from the class-conditional distribution to which the population statement applies. Dataset shift and adaptive support arrivals depending on calibration labels are outside that statement.

Appendix J.2. Weighted protection bound

Let B be the number of base classes and n_c their calibration counts. Assign each observation of class c weight $w_i = 1/(Bn_c)$, so $\sum_i w_i = 1$, and let $b = \max_i w_i$. For any fixed bounded losses $L_i \in [0, 1]$, write $q = \sum_i w_i L_i$ and $\mu = \sum_i w_i \mathbb{E}L_i$. Conditional on the score models and the independently selected threshold family, independence gives, for $t < 0$,

$$\log \mathbb{E}e^{tq} \le \sum_i \log(1 - \mu_i + \mu_i e^{tw_i}) \tag{J.1}$$

$$\le b^{-1} \sum_i w_i \log(1 - \mu_i + \mu_i e^{tb}) \tag{J.2}$$

$$\le b^{-1} \log(1 - \mu + \mu e^{tb}), \tag{J.3}$$

where $\mu_i = \mathbb{E}L_i$. The first inequality uses boundedness and the chord of the exponential; the second uses convexity of each log-moment-generating function in w_i and its zero value at zero; the third uses concavity in μ_i. Chernoff optimization yields

$$\Pr(q \le x) \le \exp[-\operatorname{kl}(x, \mu)/b], \qquad x < \mu, \tag{J.4}$$

where kl is Bernoulli relative entropy. Thus, for the null $\mu \ge \alpha$, the value

$$p(q) = \begin{cases} \exp[-\operatorname{kl}(q, \alpha)/b], & q < \alpha, \\ 1, & q \ge \alpha \end{cases} \tag{J.5}$$

is a valid lower-tail p-value. Unlike an ordinary pooled binomial calculation, this bound retains the unequal macro weights. Its dependence on the largest weight makes sparse class-conditional calibration sets conservative.

Apply the bound to the protection loss of main Eq. (8). Let $A_{\text{ref}} = \Pr(R = 1)$, $D = \Pr(R = 1, C_d = 0)$ and $G = \Pr(R = 0, C_d = 1)$. Then

$$\mathbb{E}L_d = D + (1 - r)(1 - A_{\text{ref}}), \qquad A_B(d) = A_{\text{ref}} - D + G. \tag{J.6}$$

Consequently $\mathbb{E}L_d \le 1 - r$ implies $A_B(d) \ge r A_{\text{ref}}$. The reference uncertainty is included in the paired loss rather than replaced by a point estimate. Retained base scores have $G = 0$; other readouts may have $G > 0$, so protection is sufficient and can be stricter than accuracy-ratio control for replacement models.

Appendix J.3. Threshold search and trajectory confidence

For each method and arrival, the independently fixed finite threshold family consists of unique quantiles at 0,1,. . . ,100% of base tuning margins, in

increasing order of novel admission. Following fixed-sequence testing [21], test this family in order at level $\delta/5$, stop at the first failure, and return the last accepted threshold. A false certification requires rejecting the first true null in the fixed order, whose probability is at most $\delta/5$; independence between these tests is unnecessary. A union bound over the five prespecified arrivals gives error probability at most δ for each method/order/partition. The same calibration pool may be reused because the score models, threshold families and support arrivals do not adapt to its labels. No simultaneous confidence claim across methods, orders, partitions, datasets or separately reported δ values is made.

If the accepted finite set is empty, the policy returns the original reference classifier and admits no novel class. This fallback structurally preserves its original accuracy and has zero novel accuracy. It is reported separately from useful certified expansion. The confidence levels 90,95,99% and empirical target fractions 95,97,99% were specified before the new experiment; the primary setting uses 95% retention and 95% confidence. Their complete results, worst shortfalls and five-session trajectories are saved in `independen t_risk_summary.json` and `independent_risk_summary.csv`.

Table S25: ViT-L/14 independent calibration under the original allocation in Section J. All operational means include every partition and base-only fallback; certification counts identify finite accepted thresholds. Section L expands calibration coverage with these score models frozen.

Dataset	Readout	Emp. met	Emp. novel	Cert. exp.	Test met	Cert. base	Cert. novel
CIFAR-100	Retained	7/10	76.28	0/10	10/10	88.29	0.00
CIFAR-100	Recalibrated removal	6/10	35.32	0/10	10/10	88.29	0.00
CIFAR-100	Dense prototypes	6/10	71.92	0/10	10/10	88.29	0.00
CIFAR-100	Raw fusion	5/10	34.46	0/10	10/10	88.29	0.00
CIFAR-100	Fixed temperature	8/10	71.36	0/10	10/10	88.29	0.00
CIFAR-100	Class-mean cache	4/10	65.73	0/10	10/10	88.29	0.00
Pets	Retained	1/10	93.07	0/10	10/10	96.05	0.00
Pets	Recalibrated removal	4/10	93.29	0/10	10/10	96.05	0.00
Pets	Dense prototypes	0/10	95.08	0/10	10/10	96.05	0.00
Pets	Raw fusion	4/10	86.90	0/10	10/10	96.05	0.00
Pets	Fixed temperature	3/10	91.87	0/10	10/10	96.05	0.00
Pets	Class-mean cache	1/10	92.50	0/10	10/10	96.05	0.00
ImageNet-R	Retained	7/10	83.45	10/10	10/10	89.73	77.64
ImageNet-R	Recalibrated removal	4/10	57.36	0/10	10/10	91.54	0.00
ImageNet-R	Dense prototypes	6/10	82.47	5/10	10/10	91.04	24.12
ImageNet-R	Raw fusion	3/10	64.80	4/10	10/10	91.08	26.42
ImageNet-R	Fixed temperature	5/10	81.28	10/10	10/10	89.76	65.69
ImageNet-R	Class-mean cache	4/10	79.37	2/10	10/10	91.18	6.20

Table S26: ViT-B/32 independent calibration under the original allocation. Infeasible certification triggers the shared base-only reference fallback and zero novel accuracy.

Dataset	Readout	Emp. met	Emp. novel	Cert. exp.	Test met	Cert. base	Cert. novel
CIFAR-100	Retained	1/10	60.64	0/10	10/10	79.08	0.00
CIFAR-100	Recalibrated removal	2/10	47.79	0/10	10/10	79.08	0.00
CIFAR-100	Dense prototypes	2/10	52.74	0/10	10/10	79.08	0.00
CIFAR-100	Raw fusion	2/10	50.33	0/10	10/10	79.08	0.00
CIFAR-100	Fixed temperature	5/10	55.06	0/10	10/10	79.08	0.00
CIFAR-100	Class-mean cache	2/10	57.79	0/10	10/10	79.08	0.00
Pets	Retained	1/10	75.10	0/10	10/10	90.24	0.00
Pets	Recalibrated removal	10/10	28.49	0/10	10/10	90.24	0.00
Pets	Dense prototypes	4/10	54.67	0/10	10/10	90.24	0.00
Pets	Raw fusion	6/10	53.57	0/10	10/10	90.24	0.00
Pets	Fixed temperature	5/10	64.90	0/10	10/10	90.24	0.00
Pets	Class-mean cache	7/10	42.00	0/10	10/10	90.24	0.00
ImageNet-R	Retained	2/10	56.51	10/10	10/10	73.66	45.30
ImageNet-R	Recalibrated removal	9/10	3.58	0/10	10/10	75.01	0.00
ImageNet-R	Dense prototypes	4/10	43.48	0/10	10/10	75.01	0.00
ImageNet-R	Raw fusion	9/10	3.75	0/10	10/10	75.01	0.00
ImageNet-R	Fixed temperature	2/10	52.45	7/10	10/10	74.24	27.28
ImageNet-R	Class-mean cache	5/10	33.38	0/10	10/10	75.01	0.00

Appendix K. Reproducibility details for the concise main text

Appendix K.1. Dataset construction and access

We evaluate expansion on CIFAR-100 [22] with long-tailed base supervision. The training class counts follow an exponential schedule with imbalance parameter $\rho = 100$ and nominal maximum 500 images. Five images per class are reserved before constructing the long-tailed pool, so the largest available pool contains 495 images. The remaining long-tailed pool contains 10,894 images, split into 10,487 fitting images and 407 held-out validation images. Validation reserves up to five examples per class, with smaller allocations for the rarest classes.

For seeds $\{0, \ldots, 9\}$, stratified sampling within head, medium, and tail frequency buckets yields distinct partitions of 40 base, 40 novel, and 20 unused classes. Proportional quotas use largest-remainder rounding after shuffling class membership. Evaluation uses the 80 base and novel classes.

Oxford-IIIT Pets [23] provides a complementary fine-grained task with 37 cat and dog breeds. We use its official trainval/test split and reserve five training images per breed as support. From the remaining training images, we construct an exponential long-tailed pool with $\rho = 10$ and maximum class count 90, giving 1,168 fitting and 146 validation images. Each true partition contains 15 base, 15 novel, and seven unused breeds, sampled with the same allocation rule. The official test split contains 3,669 images. Unlike CIFAR, its images have varied spatial resolutions; RGB images are resized to 224×224 by PIL bicubic interpolation before the same CLIP normalization. These features are freshly extracted for this study.

All primary expansion readouts use frozen OpenAI CLIP ViT-L/14 or

ViT-B/32 features from a single 224×224 bicubic-resized image with CLIP channel normalization. Fitting, validation, and support pools are disjoint. Supplementary Section C gives feature-cache and image-pool verification details.

Base ridge regularization is selected from $\{0.01, 0.1, 1, 10\}$ by base validation macro accuracy; the pseudo-base classifier uses the same rule. The primary comparison uses all five reserved support images, ten distinct class partitions and two arrival orders per partition. Earlier one/two/five-shot and repeated-calibration studies use three original partitions and are reported separately in Supplementary Section G.

Classifier fitting, regularization selection, and gate calibration use base classes exclusively. Five-shot CIFAR expansion consumes 200 support images for 40 novel classes. Its balanced test set supplies 4,000 base and 4,000 novel images for the expanded metrics.

ImageNet-R [20] extends the active vocabulary to 200 rendition classes: 100 base classes followed by five arrivals of 20 classes. After removing 59 byte-identical duplicates, per-class seeded splits reserve approximately 20% each for validation and test, then five support images. The fitting pool is downsampled by class rank with a nominal 100:1 schedule, yielding 2–136 images per class after availability caps. The resulting pools contain 6,438 fitting, 5,911 validation, 1,000 support and 5,911 test images. Ten frequency-stratified class partitions and two arrival orders use the same selection and reporting rules as the other datasets. RGB images are resized to 224×224 with bicubic interpolation and encoded by the same selected frozen checkpoint within each encoder comparison.

The ridge readout uses an augmented constant feature, inverse-frequency sample weights and an unregularized intercept; only the feature slopes receive ridge regularization. Class prototypes are normalized means of the permitted fitting or support features.

Appendix K.2. Pseudo-expansion selection

Fusion calibration must respect the distinction between base supervision and novel support. We construct a pseudo-expansion episode by splitting the base classes into approximately equal pseudo-base and pseudo-novel groups: 20/20 on CIFAR-100, 7/8 on Pets and 50/50 on ImageNet-R. A separate classifier is fitted using the pseudo-base classes alone. The pseudo-novel rows contain names and k support examples from those base classes, while held-out base validation images supply query labels. This episode simulates missing learned evidence without consuming any actual novel-class validation labels.

For each support budget, candidate coefficients are evaluated on this episode. The selection objective is the harmonic mean of macro accuracy on the two groups, $H = 2A_B A_N/(A_B + A_N)$ We set $H = 0$ when both terms vanish. The calibration grids are $a \in \{0, 0.25, 0.5, 1, 2\}$, $u \in \{0, 0.25, 0.5, 1, 2, 4\}$, and offsets $d \in [-4, 4]$ in steps of 0.25. Ties follow a fixed grid order. The full base classifier is then combined with actual novel rows using the selected coefficients. Neither actual novel validation queries nor test labels enter this selection rule.

This procedure uses one pseudo-class split per true partition and the same support-index seed as the evaluated support condition. CIFAR and Pets pseudo splits use seed $100 + p$ for class-partition seed p; ImageNet-R selects half the base classes with seed $900 + p$. Coefficients are chosen once for a

support budget and stay fixed when the vocabulary-size stress test appends successive subsets of novel classes.

Appendix K.3. A normalization example

A two-class example illustrates the effect. Suppose the old text scores are $(0, 2)$, giving standardized scores $(-1, 1)$, while the learned branch contributes $(0.25, -0.25)$. The second old class has the larger fused score. Appending a text score of 100 changes the two old standardized scores to approximately $(-0.728, -0.685)$. Their text separation is now about 0.043, and the unchanged learned contribution reverses the old-class ordering. No classifier weight has changed. The example isolates a normalization effect independently of whether the new class ultimately wins the expanded decision.

Appendix L. Independent calibration coverage and label budgets

Appendix L.1. Frozen models and disjoint calibration images

The locked protocol in `experiments/kbs_calibration_budget_protocol.md` fixes all score models, tuning parameters, support, test predictions and candidate thresholds from Section J before varying calibration coverage. Six readouts receive identical class-wise subsets: retained tables, recalibrated classifier removal, dense anchored prototypes, raw fusion, fixed-temperature fusion and class-mean Tip-Adapter. No score-model refitting or hyperparameter selection uses these calibration labels. Previously cached calibration scores are reused exactly after checking reconstruction against the frozen models.

The pool prioritizes Section J's independent calibration images, then adds previously unused training images up to 40 per class. Image identities and content hashes exclude fitting, tuning, support and test images. The audit

uses raw pixel hashes on CIFAR, resized-image pixel hashes on Pets, and file-content hashes on ImageNet-R. Across all classes, the resulting pools contain 4,000 CIFAR images, 1,480 Pets images and 7,376 ImageNet-R images, of which 1,805, 970 and 3,020 are supplemental. Each experiment selects its base classes from these pools. Both encoders reuse their pinned local checkpoints; only supplemental images require new encoding. Witness re-encodings verify agreement with the cached feature space.

Per-class caps $m \in \{2, 5, 10, 20, 40\}$ use nested random subsets selected without scores or correctness outcomes, with subset seeds 0, 1 and 2. At cap 40 these seeds select the same full pool, which is counted once. CIFAR and Pets have 40 available images per class; ImageNet-R has a minimum of 14, with 44 classes below 40. At cap 40 the actual base-calibration query totals are 1,600, 600 and 3,636–3,778 across partitions, respectively. The cap changes both total sample size and tail-class coverage; the curves do not isolate a causal effect of total label count alone.

Appendix L.2. A necessary coverage condition for certification

For the weighted bound in Section J, even $q = 0$ gives

$$p(0) = \exp[-\operatorname{kl}(0, 1 - r)/b] = r^{1/b}. \tag{L.1}$$

Since $p(q)$ increases on $0 \le q < 1 - r$, passing at level $\delta/5$ requires

$$n_{\text{eff}} = 1/b = B \min_c n_c \ge \frac{\log(\delta/5)}{\log r}. \tag{L.2}$$

At $r = 0.95$ and $\delta = 0.05$, this floor is 89.78. It is a necessary condition for this certificate, not a sufficient condition for successful expansion. Under the original allocation, CIFAR has $n_{\text{eff}} = 80$, Pets has 45 or 60 and ImageNet-R

has 600, 700 or 800. Thus the original CIFAR/Pets calibration allocation cannot pass this bound even at zero observed loss. Balanced per-class caps must be at least 3 for CIFAR, 6 for Pets and 1 for ImageNet-R to clear the zero-loss floor. Actual protection losses require further evidence. A balanced allocation can improve the bound even with fewer total labels than a larger, uneven pool.

Appendix L.3. Aggregation, admission and complete comparisons

Main Fig. 2 and Fig. S6 show the fifth arrival in order 0, the first prespecified order. Novel-accuracy means average three subset seeds within each of ten partitions below cap 40, and use the full pool once per partition at 40. Retained-table shading shows sample standard deviation across ten partition means, clipped to [0,100]. Finite-admission fractions count 30 correlated partition/subset cells below 40 and ten at 40, with equal partition weights; these cells are not independent Bernoulli trials. Confidence applies to five arrivals for each prespecified method/order/partition and budget separately, without a simultaneous statement across plotted budgets or methods.

All six retained-table dataset/encoder conditions certify final expansion in 10/10 partitions and meet all ten final test targets. Across both orders, all 20 paths per condition meet the test target at every arrival. Finite admission throughout holds for all CIFAR and ImageNet-R paths, and for 17/20 Pets paths with L/14 and 16/20 with B/32; other Pets paths use an early fallback. A path containing fallback therefore counts toward test protection but not uninterrupted useful admission.

Paired policy differences include all ten partitions, including fallback; `calibration_budget_paired.csv` additionally reports jointly finite partition

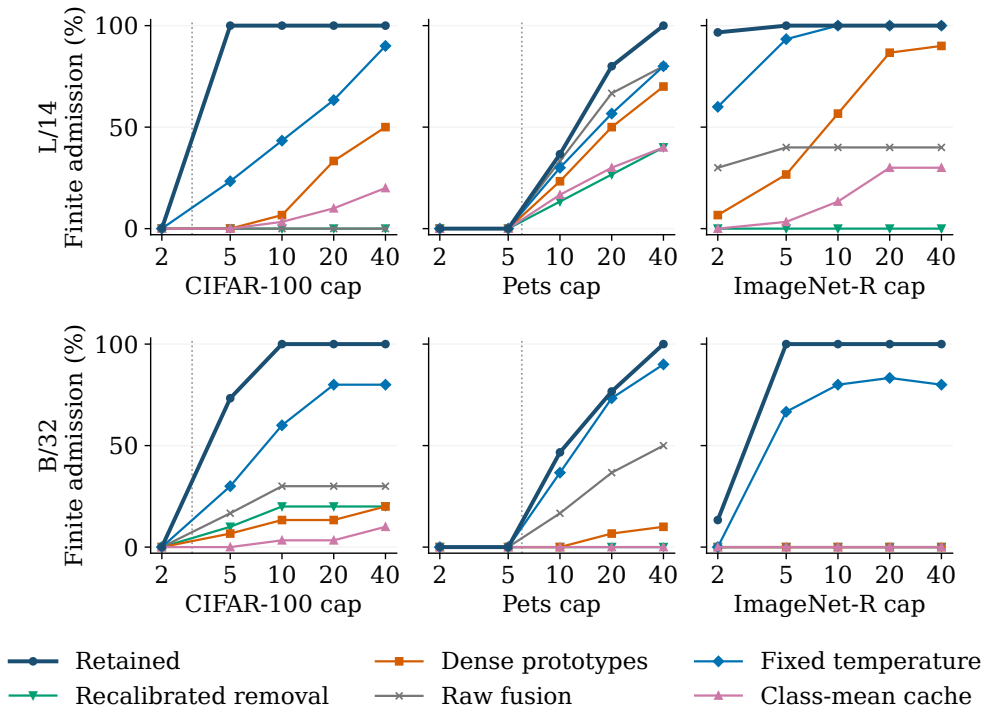


Figure S6: Finite certified admission after the fifth arrival in order 0, under 95% retention and 95% five-arrival confidence. Fractions use 30 partition/subset cells below cap 40 and ten full-pool cells at 40. Grey lines mark the necessary balanced cap for the zero-loss bound; ImageNet-R's floor of one lies below the plotted range. Every readout receives the same images and protection rule. Base-only fallback counts as no finite admission.

Table S27: Complete L/14 cap-40 comparison after the fifth arrival in order 0. Empirical thresholds and independent risk certification use the same frozen score models and calibration pools. All means include ten partitions, with zero novel accuracy on base-only fallback. Exp. counts finite admissions; Test met counts test targets.

Dataset	Readout	Emp. met	Emp. novel	Cert. exp.	Test met	Base	Novel
CIFAR-100	Retained	8/10	76.90	10/10	10/10	85.86	72.76
CIFAR-100	Recalibrated removal	6/10	47.00	0/10	10/10	88.29	0.00
CIFAR-100	Dense prototypes	6/10	72.93	5/10	10/10	87.45	30.74
CIFAR-100	Raw fusion	6/10	43.13	0/10	10/10	88.29	0.00
CIFAR-100	Fixed temperature	6/10	71.75	9/10	10/10	86.66	49.14
CIFAR-100	Class-mean cache	5/10	66.31	2/10	10/10	87.94	9.35
Pets	Retained	4/10	92.58	10/10	10/10	94.10	88.51
Pets	Recalibrated removal	7/10	92.62	4/10	10/10	95.48	32.68
Pets	Dense prototypes	3/10	94.54	7/10	10/10	94.93	62.39
Pets	Raw fusion	6/10	86.52	8/10	10/10	94.91	60.83
Pets	Fixed temperature	5/10	91.51	8/10	10/10	95.02	69.61
Pets	Class-mean cache	4/10	91.84	4/10	10/10	95.53	34.86
ImageNet-R	Retained	8/10	83.53	10/10	10/10	89.02	80.74
ImageNet-R	Recalibrated removal	4/10	54.87	0/10	10/10	91.54	0.00
ImageNet-R	Dense prototypes	7/10	82.65	9/10	10/10	90.44	48.58
ImageNet-R	Raw fusion	4/10	64.01	4/10	10/10	90.84	27.88
ImageNet-R	Fixed temperature	5/10	81.38	10/10	10/10	89.19	76.00
ImageNet-R	Class-mean cache	5/10	79.08	3/10	10/10	90.99	10.31

Table S28: Complete B/32 cap-40 comparison, with access and fallback accounting as in the L/14 table.

Dataset	Readout	Emp. met	Emp. novel	Cert. exp.	Test met	Base	Novel
CIFAR-100	Retained	1/10	59.19	10/10	10/10	76.74	52.54
CIFAR-100	Recalibrated removal	7/10	34.79	2/10	10/10	78.57	10.73
CIFAR-100	Dense prototypes	6/10	51.14	2/10	10/10	78.79	7.95
CIFAR-100	Raw fusion	5/10	37.98	3/10	10/10	78.38	13.51
CIFAR-100	Fixed temperature	4/10	55.06	8/10	10/10	77.87	36.50
CIFAR-100	Class-mean cache	3/10	54.52	1/10	10/10	79.21	4.16
Pets	Retained	3/10	73.15	10/10	10/10	88.10	61.94
Pets	Recalibrated removal	10/10	35.20	0/10	10/10	90.24	0.00
Pets	Dense prototypes	5/10	54.36	1/10	10/10	90.17	5.89
Pets	Raw fusion	6/10	56.09	5/10	10/10	90.03	17.69
Pets	Fixed temperature	4/10	65.62	9/10	10/10	90.46	41.26
Pets	Class-mean cache	8/10	55.72	0/10	10/10	90.24	0.00
ImageNet-R	Retained	3/10	56.13	10/10	10/10	72.98	49.67
ImageNet-R	Recalibrated removal	10/10	0.00	0/10	10/10	75.01	0.00
ImageNet-R	Dense prototypes	3/10	47.78	0/10	10/10	75.01	0.00
ImageNet-R	Raw fusion	10/10	0.00	0/10	10/10	75.01	0.00
ImageNet-R	Fixed temperature	4/10	52.37	8/10	10/10	73.68	33.60
ImageNet-R	Class-mean cache	4/10	32.26	0/10	10/10	75.01	0.00

counts and conditional accuracy differences. These distinguish improved ability to certify admission from predictive differences after both methods admit. For example, L/14 retained-minus-dense policy differences are 42.02 [23.83,60.07] points on CIFAR and 26.12 [−0.01,53.10] on Pets. On their jointly finite sets, the means are 12.84 points over five CIFAR partitions and −0.63 over seven Pets partitions. Conditional paired percentile intervals resample ten partition means 20,000 times with seed 2026; they describe class-assignment variation on fixed images. The sufficient protection criterion ignores recovery of reference errors by replacement models, as derived in Section J.

The 93,600 records cover 60 dataset/encoder/partition runs, two orders, five arrivals, 13 distinct cap/subset settings, six methods and two policies. The independent verifier reconstructs 88,216 threshold tests, selected thresholds and final predictions; it verifies zero protected-pool overlap and unchanged original models and test archives. A completed rerun preserves hashes and modification times of 1,996 checkpoint files. Complete cap-specific metrics, query counts and trajectories are supplied in `calibration_budget_summary.json` and its CSV export. The study uses previously examined benchmarks; the population confidence statement retains Section J's class-conditional sampling assumptions.

References

[1] M. D. McDonnell, D. Gong, A. Parvaneh, E. Abbasnejad, A. van den Hengel, RanPAC: Random Projections and Pre-trained Models for Continual Learning, in: Advances in Neural Information Processing Systems, Vol. 36, 2023, pp. 12022–12053. doi:10.52202/075280-0526.
URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/2793dc35e14003dd367684d93d236847-Abstract-Conference.html

[2] K. Cao, C. Wei, A. Gaidon, N. Arechiga, T. Ma, Learning Imbalanced Datasets with Label-Distribution-Aware Margin Loss, arXiv preprint arXiv:1906.07413 (2019).
URL https://arxiv.org/abs/1906.07413

[3] Y. Cui, M. Jia, T.-Y. Lin, Y. Song, S. Belongie, Class-Balanced Loss Based on Effective Number of Samples, arXiv preprint arXiv:1901.05555 (2019).
URL https://arxiv.org/abs/1901.05555

[4] J. Ren, C. Yu, S. Sheng, X. Ma, H. Zhao, S. Yi, H. Li, Balanced Meta-Softmax for Long-Tailed Visual Recognition, arXiv preprint arXiv:2007.10740 (2020).
URL https://arxiv.org/abs/2007.10740

[5] A. K. Menon, S. Jayasumana, A. S. Rawat, H. Jain, A. Veit, S. Kumar, Long-tail learning via logit adjustment, arXiv preprint arXiv:2007.07314 (2020).
URL https://arxiv.org/abs/2007.07314

[6] B. Kang, S. Xie, M. Rohrbach, Z. Yan, A. Gordo, J. Feng, Y. Kalantidis, Decoupling Representation and Classifier for Long-Tailed Recognition, arXiv preprint arXiv:1910.09217 (2019).
URL https://arxiv.org/abs/1910.09217

[7] J.-X. Shi, T. Wei, Z. Zhou, J.-J. Shao, X.-Y. Han, Y.-F. Li, Long-Tail Learning with Foundation Model: Heavy Fine-Tuning Hurts, in: International Conference on Machine Learning, 2024, pp. 45014–45039.
URL https://proceedings.mlr.press/v235/shi24g.html

[8] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, W. Chen, LoRA: Low-Rank Adaptation of Large Language Models, arXiv preprint arXiv:2106.09685 (2021).
URL https://arxiv.org/abs/2106.09685

[9] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, N. Houlsby, An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale, arXiv preprint arXiv:2010.11929 (2020).
URL https://arxiv.org/abs/2010.11929

[10] Z. Zhong, J. Cui, S. Liu, J. Jia, Improving Calibration for Long-Tailed Recognition, arXiv preprint arXiv:2104.00466 (2021).
URL https://arxiv.org/abs/2104.00466

[11] T. Wei, J.-X. Shi, M.-L. Zhang, Y.-F. Li, Robust long-tailed learning under label noise, Frontiers of Computer Science 20 (1) (2026) 2001321. doi:10.1007/s11704-025-40860-0.
URL https://journal.hep.com.cn/fcs/EN/10.1007/s11704-025-40860-0

[12] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, I. Sutskever, Learning Transferable Visual Models From Natural Language Supervision, in: International Conference on Machine Learning, 2021, pp. 8748–8763.
URL https://proceedings.mlr.press/v139/radford21a.html

[13] K. Zhou, J. Yang, C. C. Loy, Z. Liu, Learning to Prompt for Vision-Language Models, International Journal of Computer Vision 130 (9) (2022) 2337–2348. doi:10.1007/s11263-022-01653-1.
URL https://link.springer.com/article/10.1007/s11263-022-01653-1

[14] K. Zhou, J. Yang, C. C. Loy, Z. Liu, Conditional Prompt Learning for Vision-Language Models, arXiv preprint arXiv:2203.05557 (2022).
URL https://arxiv.org/abs/2203.05557

[15] J. Snell, K. Swersky, R. S. Zemel, Prototypical Networks for Few-shot Learning, arXiv preprint arXiv:1703.05175 (2017).
URL https://arxiv.org/abs/1703.05175

[16] Y. Wang, W.-L. Chao, K. Q. Weinberger, L. van der Maaten, SimpleShot: Revisiting Nearest-Neighbor Classification for Few-Shot Learning, arXiv preprint arXiv:1911.04623 (2019).
URL https://arxiv.org/abs/1911.04623

[17] R. Zhang, W. Zhang, R. Fang, P. Gao, K. Li, J. Dai, Y. Qiao, H. Li, Tip-Adapter: Training-free Adaption of CLIP for Few-shot Classification, arXiv preprint arXiv:2207.09519 (2022).
URL https://arxiv.org/abs/2207.09519

[18] Y. Zhu, F. Zhuang, X. Zhang, Z. Qi, Z. Shi, J. Cao, Q. He, Combat data shift in few-shot learning with knowledge graph, Frontiers of Computer Science 17 (1) (2023) 171305. doi:10.1007/s11704-022-1339-7.
URL https://journal.hep.com.cn/fcs/EN/10.1007/s11704-022-1339-7

[19] K. Książek, H. Jastrzębski, K. Pniaczek, B. Trojan, M. Karp, J. Tabor, FeNeC: Enhancing continual learning via feature clustering with neighbor- or logit-based classification, Knowledge-Based Systems 338 (2026) 115479. doi:10.1016/j.knosys.2026.115479.

[20] D. Hendrycks, S. Basart, N. Mu, S. Kadavath, F. Wang, E. Dorundo, R. Desai, T. Zhu, S. Parajuli, M. Guo, D. Song, J. Steinhardt, J. Gilmer, The Many Faces of Robustness: A Critical Analysis of Out-of-Distribution Generalization, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 8340–8349.
URL https://openaccess.thecvf.com/content/ICCV2021/html/Hendrycks_The_Many_Faces_of_Robustness_A_Critical_Analysis_of_Out-of-Distribution_ICCV_2021_paper.html

[21] A. N. Angelopoulos, S. Bates, E. J. Candès, M. I. Jordan, L. Lei, Learn then test: Calibrating predictive algorithms to achieve risk control, arXiv preprint arXiv:2110.01052 (2021). doi:10.48550/arXiv.2110.01052.

[22] A. Krizhevsky, Learning multiple layers of features from tiny images, Tech. rep., University of Toronto (2009).
URL https://www.cs.toronto.edu/~kriz/learning-features-2009-TR.pdf

[23] O. M. Parkhi, A. Vedaldi, A. Zisserman, C. V. Jawahar, Cats and dogs, in: IEEE Conference on Computer Vision and Pattern Recognition, 2012, pp. 3498–3505. doi:10.1109/CVPR.2012.6248092.
URL https://www.robots.ox.ac.uk/~vgg/publications/2012/parkhi12a/